

CANONICAL NILPOTENT SPECIALIZATION, STRATIFIED CONTACT ALGEBRA, AND POLARIZED RECONSTRUCTION IN MULTIPLICATION FAILURE

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ABSTRACT. We study the nonreduced Fitting scheme on a Grassmannian on which multiplication fails for cube-zero algebras with Hilbert function $(1, 4, 6)$. The first nilpotent layer reconstructs the ramification quartic K3 surface together with its quartic polarization. On the whole Grassmannian the nilpotent algebra is controlled by a canonical residual-colon filtration. If

$$\mathcal{J}^{\text{res}} = (\mathcal{I}_{\widehat{D}} : \mathcal{I}_{\widehat{\Delta}}), \quad \mathcal{L} = \mathcal{I}_{\widehat{\Delta}} / \mathcal{I}_{\widehat{\Delta}}^2,$$

then

$$\mathcal{N}^j / \mathcal{N}^{j+1} \simeq \mathcal{L}^{\otimes j} \otimes \mathcal{O}_{W_j}.$$

Revision 126 upgrades this graded description to a canonical specialization theorem. The extended Rees algebra

$$\mathcal{R}_{\mathcal{N}} = \mathcal{O}_{\widehat{D}}[t, \mathcal{N}t^{-1}] \subset \mathcal{O}_{\widehat{D}}[t, t^{-1}]$$

is flat over $\mathbb{A}^1$; its generic fibre is the original multiplication-failure scheme and its special fibre is the intrinsic normal cone

$$C_{\widehat{\Delta}/\widehat{D}} = \text{Spec}_{\widehat{\Delta}} \text{gr}_{\mathcal{N}} \mathcal{O}_{\widehat{D}}.$$

Thus all projection coranks and rank-drop loci occur in one functorial degeneration rather than as unrelated local normal forms.

Across the first singular Schubert boundary we compute that special fibre explicitly. On the mixed-regular projection-rank-two locus,

$$\mathcal{I}_{\widehat{D}} = \delta I_{\text{ct}} \cap \mathfrak{m}^5.$$

The vertex factor remains an actual primary component through contact collisions; a contact of multiplicity e has generic primary type

$$(\delta^2, \delta q^e, q^{e+1}).$$

If the restricted quartic vanishes identically, the exact local ideal is $\delta^2 \mathfrak{m} = (\delta^2) \cap \mathfrak{m}^5$. Crossing the first decomposable mixed-kernel wall changes the second intrinsic depth support to a canonical length-three nonreduced thickening.

The same abstract failure scheme reconstructs the polarized quartic K3. The classical Reye/nodal-Enriques geometry accounts for the nine-dimensional source quotient, while the nonclassical theorem is the intrinsic polarized reconstruction together with the global Rees/normal-cone specialization and its explicit singular-boundary algebra. On dense opens the remaining Torelli ambiguity is a finite etale cover with transitive geometric monodromy.

1. INTRODUCTION AND STATEMENTS

A degeneracy locus remembers where a map loses rank; its scheme structure can remember substantially more. In the multiplication problem studied here, the nilpotent layers of a Fitting scheme recover geometric data that disappear after reduction. For Hilbert function $(1, 4, 6)$, the first layer recovers a polarized quartic K3 surface. The next layer detects the first singular

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Schubert stratum. The purpose of this revision is to determine how those two layers fit together across the rank boundary, including contact collisions and the first failure of the mixed-regular hypothesis.

Let V be a complex vector space of dimension $e \in \{3, 4\}$ and put $p = e(e-1)/2$. For an e -dimensional relation space

$$R \subset \text{Sym}^2 V$$

set

$$S_R = \text{Sym}^2 V / R, \quad B_R = \mathbb{C} \oplus V \oplus S_R,$$

where multiplication on V is the quotient map and $(V \oplus S_R)S_R = 0$. On

$$X_R = \text{Gr}(e, V \oplus S_R)$$

let $\mathcal{K}$ be the tautological subbundle and consider

$$\hat{D}_R = V(\text{Fitt}_0 \text{ coker}[\text{Sym}^m(\mathcal{O} \oplus \mathcal{K}) \longrightarrow B_R \otimes \mathcal{O}]), \quad m \geq 2. \quad (1.1)$$

The scheme is independent of m . Its reduction is the Schubert divisor $\hat{\Delta}_R$ on which $\mathcal{K} \rightarrow V$ drops rank.

Regard R as a web of quadrics on $\mathbb{P}(V^*)$. On the nonempty invariant open G_e° used below the web is basepoint-free and the Jacobian hypersurface

$$Y_R = V(\det(\partial q_i / \partial x_j)) \subset \mathbb{P}(V^*) \quad (1.2)$$

is smooth. It is a genus-one curve for $e = 3$ and a quartic K3 surface for $e = 4$.

The first nilpotent layer. Let $U_R \subset X_R$ be the projection-corank-at-most-one open and write $D_R = \hat{D}_R \cap U_R$. The local normal form on this open is

$$\mathcal{I}_{D_R} = t(t, f) = (t) \cap (t, f)^2,$$

where $t = 0$ is the Schubert divisor and $f = 0$ is the ramification equation on it. If $\mathcal{N}_R$ is the nilradical of $\mathcal{O}_{D_R}$, then

$$\mathcal{N}_R^2 = 0, \quad \text{Ann}(\mathcal{N}_R) = \mathcal{I}_{E_R} / \mathcal{I}_{D_R},$$

and E_R is a smooth intrinsic scheme.

Theorem 1.1 (Polarized reconstruction). *For $R \in G_e^\circ$, put*

$$\mathcal{L}_R = \omega_{E_R} \otimes \mathcal{N}_R^{\otimes -(p+e-1)}.$$

There is an isomorphism of graded algebras

$$\bigoplus_{n \geq 0} H^0(E_R, \mathcal{L}_R^{\otimes n}) \simeq \bigoplus_{n \geq 0} H^0(Y_R, \mathcal{O}_{Y_R}(en)).$$

Consequently D_R determines $(Y_R, \mathcal{O}_{Y_R}(e))$. For $e = 4$ it determines $(Y_R, \mathcal{O}_{Y_R}(1))$. The same conclusions hold with $\hat{D}_R$ in place of D_R ; no isomorphism is required to preserve the ambient Grassmannian, the relation space, or any chosen projection.

The nilpotent filtration across projection rank. Let $\widehat{\mathcal{N}}_R$ denote the nilradical of $\mathcal{O}_{\widehat{D}_R}$. For $j \geq 1$ define

$$\mathcal{Z}_j(\widehat{D}_R) = V\left(\text{Ann}_{\mathcal{O}_{\widehat{D}_R}}(\widehat{\mathcal{N}}_R^j)\right). \quad (1.3)$$

These schemes are intrinsic and satisfy $\mathcal{Z}_{j+1} \subset \mathcal{Z}_j$. The global theorem below identifies the entire associated graded nilpotent algebra on the whole Grassmannian by residual colon ideals. Its subsequent clauses compute the first two nonzero pieces geometrically across the projection-corank-one/projection-corank-two boundary and through the first tensor-rank wall.

At a projection-rank-two point let $H \subset V$ be the image and $W = V/H$. Write

$$A_H : \text{Sym}^2 H \rightarrow S_R$$

for the pure H -block and

$$\overline{B}_H : H \otimes W \longrightarrow S_R / \text{im } A_H$$

for the induced mixed block. We call (R, H) *mixed-regular* when A_H is injective, $\overline{B}_H$ is surjective, and its one-dimensional kernel is generated by a rank-two tensor. This condition does not impose simple contact roots.

On a mixed-regular transverse slice write

$$T = \begin{pmatrix} a & b \\ c & d \end{pmatrix}, \quad \delta = ad - bc, \quad \mathbf{m} = (a, b, c, d).$$

The restricted quartic $Y_R \cap \mathbb{P}(H^\perp)$ is represented by a binary quartic h_H . The root-free residual calculation defines a canonical unsplit contact ideal

$$I_H = (\delta, g_0, \dots, g_4),$$

whose projectivization on the rank-one Segre quadric is the Cartier divisor

$$V(h_H) \times \mathbb{P}^1.$$

Theorem 1.2 (Stratified nilpotent algebra at the first singular Schubert boundary). *Let $R \in G_4^\circ$.*

(1) *On the smooth locus of $(\widehat{D}_R)_{\text{red}}$,*

$$\mathcal{Z}_1(\widehat{D}_R) = E_R, \quad \widehat{\mathcal{N}}_R|_{E_R} \simeq \mathcal{O}_{E_R}(-\widehat{\Delta}_R),$$

and Theorem 1.1 reconstructs the polarized quartic K3 from this first layer.

(2) *On the whole Grassmannian set*

$$\mathcal{J}_R^{\text{res}} = (\mathcal{I}_{\widehat{D}_R} : \mathcal{I}_{\widehat{\Delta}_R}), \quad \mathcal{L} = \mathcal{I}_{\widehat{\Delta}_R} / \mathcal{I}_{\widehat{\Delta}_R}^2,$$

and, for $j \geq 1$,

$$W_j = V_{\widehat{\Delta}_R} \left((\mathcal{J}_R^{\text{res}} : \mathcal{I}_{\widehat{\Delta}_R}^{j-1}) + \mathcal{I}_{\widehat{\Delta}_R} \right).$$

Then

$$\text{gr}_{\widehat{\mathcal{N}}_R} \mathcal{O}_{\widehat{D}_R} \simeq \mathcal{O}_{\widehat{\Delta}_R} \oplus \bigoplus_{j \geq 1} \mathcal{L}^{\otimes j} \otimes \mathcal{O}_{W_j}, \quad (1.4)$$

and

$$\text{Ann}(\widehat{\mathcal{N}}_R^j) = \frac{(\mathcal{J}_R^{\text{res}} : \mathcal{I}_{\widehat{\Delta}_R}^{j-1})}{\mathcal{I}_{\widehat{D}_R}}. \quad (1.5)$$

The sum is finite. These identities hold at every projection rank and every rank-drop point. On a Schur chart $\mathcal{I}_{\widehat{\Delta}_R} = (d)$, $\mathcal{I}_{\widehat{D}_R} = dJ$, they become

$$\text{Ann}(N^j) = (J : d^{j-1}) / (dJ), \quad N^j / N^{j+1} \simeq A / ((d) + (J : d^{j-1})),$$

and the exact nilpotency index is

$$1 + \min\{k \geq 0 : d^k \in J\}.$$

- (3) On every mixed-regular projection-rank-two chart,

$$\mathcal{I}_{\widehat{D}_R} = \delta I_H \cap \mathfrak{m}^5. \quad (1.6)$$

The factor $\mathfrak{m}^5$ is the actual irredundant $\mathfrak{m}$ -primary component throughout the collision boundary.

If

$$h_H = \prod_{\nu=1}^s \ell_\nu^{e_\nu}, \quad \sum_{\nu=1}^s e_\nu = 4,$$

the associated primes are

$$(\delta), P_1, \dots, P_s, \mathfrak{m},$$

where P_ν is the contact prime corresponding to ℓ_ν . At the generic point of P_ν , with $P_\nu = (\delta, q_\nu)$, the contact-primary type is

$$(\delta^2, \delta q_\nu^{e_\nu}, q_\nu^{e_\nu+1}). \quad (1.7)$$

Thus repeated contacts are controlled scheme-theoretically without choosing four distinct roots.

- (4) On the same mixed-regular neighbourhood,

$$\widehat{\mathcal{N}}_R^3 = 0 \neq \widehat{\mathcal{N}}_R^2,$$

and, writing $\mathcal{L}$ for the conormal line of the Schubert divisor and $\mathcal{C}_H = V(I_H)$ for the affine contact cone,

$$\mathrm{gr}_{\widehat{\mathcal{N}}_R} \mathcal{O}_{\widehat{D}_R} \simeq \mathcal{O}_{\widehat{\Delta}_R} \oplus (\mathcal{L} \otimes \mathcal{O}_{\mathcal{C}_H}) \oplus (\mathcal{L}^{\otimes 2} \otimes \mathcal{O}_{\Sigma_R}). \quad (1.8)$$

In particular

$$\mathcal{Z}_1 = V(I_H \cap \mathfrak{m}^3), \quad \mathcal{Z}_2 = \Sigma_R$$

scheme-theoretically. On the adjacent projection-corank-one locus, $\mathcal{Z}_1$ is the ramification support E_R ; hence (1.8) describes the specialization of the first layer to the rank-two boundary and identifies the second layer there.

- (5) If $h_H \equiv 0$, equivalently the pencil $\mathbb{P}(H^\perp)$ is a line contained in Y_R , then

$$I_H = (\delta), \quad \mathcal{I}_{\widehat{D}_R} = \delta^2 \mathfrak{m} = (\delta^2) \cap \mathfrak{m}^5. \quad (1.9)$$

The containment locus in $\mathrm{Gr}(2, V)$ is canonically $F_1(Y_R)$, the finite reduced Fano scheme of lines on the smooth quartic.

- (6) Suppose A_H is injective and $\overline{B}_H$ remains surjective with one-dimensional kernel, but that kernel becomes a decomposable tensor. Under the transverse nonvanishing condition of Definition 9.4, the nilpotent index is still three while

$$\mathcal{Z}_2 = V(c, d, a^2, ab, b^2).$$

Thus the second intrinsic depth support becomes a length-three nonreduced thickening of the rank-two vertex on the first non-mixed-regular tensor-rank wall.

The global residual-colon formulas (1.4)–(1.5) are the all-corank structure theorem. Independently, the full Fitting ideal also has the block presentation of Theorem 8.1. The block presentation is used to compute the colon ideals; it is not itself substituted for their structure.

The global part of the theorem is Theorem 9.1; the geometric identification of its first pieces is proved in Sections 4–9. Thus the paper now has a canonical all-corank associated-graded object even on rank-drop loci where no closed-form primary decomposition is chosen. On the mixed-regular rank-two boundary its central identity is

$$J = I_H \cap \mathfrak{m}^3, \quad \delta J = \delta I_H \cap \mathfrak{m}^5.$$

The first equality identifies the contact saturation hidden in the root-free residual minors; the second converts it into a genuine primary statement at the rank-two vertex. It also explains why a collision alters the contact-primary type but leaves the vertex component fixed. The associated-graded formula (1.8) then follows by cancelling successive powers of δ .

The result separates three levels of information without leaving a gap in the intrinsic filtration. The residual-colon filtration and its associated graded algebra hold on the entire Grassmannian. The residual block presentation is a computational presentation of that global object. The contact-primary specialization gives an explicit geometric identification of its first two graded supports on the entire mixed-regular rank-two locus, including repeated roots and the containment locus; the decomposable-kernel theorem then computes the first tensor-rank wall beyond mixed regularity.

The contact incidence. Put $G = \mathrm{Gr}(2, V)$, with universal quotient bundle $\mathcal{Q}$. The quartic equation of Y_R restricts to a global section

$$h_R \in H^0(G, \mathrm{Sym}^4 \mathcal{Q}).$$

The incidence

$$\mathcal{R}_R = \{(H, [\alpha]) \in G \times Y_R : H \subset \ker \alpha\}$$

is a $\mathbb{P}^2$ -bundle over Y_R , hence smooth and irreducible. Over the locus where h_H is not identically zero it is finite flat of degree four over G . Its global discriminant is

$$\mathrm{Disc}(h_R) \in H^0(G, (\det \mathcal{Q})^{12}),$$

so the discriminant divisor has Pluecker class $12H$. The interpretation of the four etale sheets as the four simple-contact primary branches requires, in addition, the mixed-regular rank conditions. On the containment locus the fibre is a projective line and the finite-cover interpretation is replaced by Theorem 9.6.

Simple contacts as the split form. On the simple-contact mixed-regular open the unsplit theorem recovers the earlier etale primary decomposition.

Theorem 1.3 (Corank-two primary structure on the simple-contact open). *There is a nonempty invariant open $G_4^\dagger \subset G_4^\circ$ such that, for each $R \in G_4^\dagger$, a dense open set of rank-two images $H \subset V$ has the following property. Near every plane with image H , after an etale extension and smooth auxiliary coordinates,*

$$\mathcal{I}_{\widehat{D}_R} = (\delta) \cap P_1^2 \cap P_2^2 \cap P_3^2 \cap P_4^2 \cap \mathfrak{m}^5. \quad (1.10)$$

The rank-two stratum is an additional associated support,

$$\widehat{\mathcal{N}}_R^3 = 0 \neq \widehat{\mathcal{N}}_R^2, \quad \mathrm{Ann}(\widehat{\mathcal{N}}_R^2) = \mathfrak{m}/\mathcal{I}_{\widehat{D}_R}.$$

Theorem 1.2 is the unsplit and stratified extension: simple roots give $e_\nu = 1$ in (1.7); collisions give higher e_ν ; and $h_H \equiv 0$ gives the separate containment algebra (1.9).

Variation and finite ambiguity. The polarized K3s vary nontrivially.

Theorem 1.4 (Variation and finite ambiguity). *For $e = 4$, the image of*

$$R \longmapsto (Y_R, \mathcal{O}_{Y_R}(1))$$

in quartic K3 moduli has dimension nine. On a nonempty invariant open, the induced map from projective equivalence classes of relation spaces to its image is generically finite. Thus, over a general point of this image, D_R and $\widehat{D}_R$ determine at most finitely many isomorphism classes of algebras B_R .

The exact rank-24 differential witness in Section 6 proves this assertion for the actual Jacobian quartic map. Section 12 separates that arithmetic witness from the classical geometric reason for the dimension count. A regular web gives the classical Reye Enriques surface and its K3 cover. The web parameter is 24-dimensional and the infinitesimal projective orbit is 15-dimensional, leaving the classical nine-dimensional quotient. We make the parameter-to-Hilbert bridge explicit and isolate the finite involution ambiguity on a fixed polarized K3.

The number nine is therefore not used as a novelty signal. The intrinsic statement is that an abstract nonreduced multiplication scheme recovers the polarized K3 and, in addition, the stratified nilpotent algebra described in Theorem 1.2.

Classical and priority boundaries. The bilinear incidence K3, the fixed-point-free factor-exchange involution, and the associated Reye congruence are classical. We compare them with Cossec, Arrondo–Sols, Dolgachev–Reider, Ingalls–Kuznetsov, and the modern nodal-Enriques treatment in Section 12. We likewise compare the nilpotent filtration with ribbons and multiple structures. The claim is not that nilpotent filtrations in general are new; it is that the particular Fitting algebra in (1.1) has intrinsic graded pieces that recover the polarized K3, the unsplit contact Cartier geometry, and the first singular Schubert support.

Ballico’s failure-locus work is treated as a genuine predecessor. The accessible 1996 continuation is compared at theorem level. The 1993 article is cited explicitly and is not used as a premise in any non-anticipation argument. The mathematical theorems of this paper are stated and proved independently of that documentary question.

Canonical specialization of the whole nonreduced scheme. The associated graded algebra in Theorem 1.2 is not only a bookkeeping device. It is the special fibre of a canonical flat degeneration of the entire multiplication-failure scheme.

Theorem 1.5 (Canonical Rees specialization). *Let $R \in G_4^\circ$, let $\widehat{\mathcal{N}}_R \subset \mathcal{O}_{\widehat{D}_R}$ be the nilradical, and form the extended Rees algebra*

$$\mathcal{R}_R = \mathcal{O}_{\widehat{D}_R}[t, \widehat{\mathcal{N}}_R t^{-1}] \subset \mathcal{O}_{\widehat{D}_R}[t, t^{-1}].$$

The affine-local spectra of $\mathcal{R}_R$ glue to a canonical scheme

$$\mathfrak{D}_R \longrightarrow \mathbb{A}^1$$

that is flat over $\mathbb{A}^1$. Over $\mathbb{G}_m$,

$$\mathfrak{D}_R|_{\mathbb{G}_m} \simeq \widehat{D}_R \times \mathbb{G}_m,$$

whereas the fibre over 0 is

$$\mathfrak{D}_{R,0} \simeq \mathrm{Spec} \widehat{\Delta}_R \operatorname{gr}_{\widehat{\mathcal{N}}_R} \mathcal{O}_{\widehat{D}_R} = C_{\widehat{\Delta}_R/\widehat{D}_R}.$$

Thus the special fibre is the intrinsic normal cone of the reduction inside the multiplication-failure scheme. By Theorem 9.1, its graded structure is

$$\mathcal{O}_{\widehat{\Delta}_R} \oplus \bigoplus_{j \geq 1} \mathcal{L}^{\otimes j} \otimes \mathcal{O}_{W_j},$$

with multiplication induced by the residual-colon tower.

The construction is functorial under abstract isomorphisms of multiplication-failure schemes. It therefore requires neither a chosen Grassmannian embedding nor a chosen projection chart.

The proof is given in Section 10. On the mixed-regular rank-two boundary its special fibre is the three-step graded algebra of Theorem 9.3; at the decomposable mixed-kernel wall the degree-two support becomes the length-three scheme of Theorem 9.5. Hence the collision and tensor-rank boundary calculations are restrictions of one global specialization mechanism, not independent normal-form assertions.

2. QUADRATIC RELATIONS AND A RELATIVE RAMIFICATION FAMILY

Put $N = e(e+1)/2$ and $P = \mathbb{P}(V^*)$. Let $\mathcal{H} \subset V \otimes \mathcal{O}_P$ be the tautological hyperplane bundle. Contraction is normalized by $c_\alpha(vw) = \alpha(v)w + \alpha(w)v$.

Lemma 2.1 (Restriction and contraction). *There is a locally split exact sequence*

$$0 \longrightarrow \mathrm{Sym}^2 \mathcal{H} \longrightarrow \mathrm{Sym}^2 V \otimes \mathcal{O}_P \xrightarrow{c} V \otimes \mathcal{O}_P(1) \longrightarrow 0.$$

The cokernels of $\mathrm{Sym}^2 \mathcal{H} \rightarrow S_R \otimes \mathcal{O}_P$ and $R \otimes \mathcal{O}_P \rightarrow V \otimes \mathcal{O}_P(1)$ are naturally isomorphic. Their determinant ideals define the same degree- e hypersurface when nonzero.

Proof. Locally choose ℓ with $\alpha(\ell) = 1$. The decomposition $\mathrm{Sym}^2 V = \mathrm{Sym}^2 H \oplus H\ell \oplus \mathbb{C}\ell^2$ identifies contraction with zero, the identity, and multiplication by two on the three summands, respectively. The twist accounts for rescaling α . Both cokernels are the quotient by $R + \mathrm{Sym}^2 \mathcal{H}$; hence their zeroth Fitting ideals agree. Finally, $\det c_R$ is a section of $\det V \otimes (\det R)^* \otimes \mathcal{O}_P(e)$. $\square$

Proposition 2.2. *If R is basepoint-free, its quadratic morphism $\phi_R : P \rightarrow \mathbb{P}(R^*)$ is finite and surjective of degree 2^{e-1} . Its ramification Fitting scheme is Y_R in (1.2). Moreover,*

$$R \text{ is basepoint-free} \iff \gamma_R(HV) = S_R \text{ for every hyperplane } H \subset V. \quad (2.1)$$

Proof. The pullback of $\mathcal{O}(1)$ is $\mathcal{O}_P(2)$. A positive-dimensional projective fibre contains a curve on which this ample line bundle would be trivial. Thus the proper morphism is quasi-finite, and hence finite. Its image has dimension $e-1$ and is the whole target. Intersecting $e-1$ pulled-back hyperplanes computes its degree.

For $J = (\partial q_i / \partial x_j)$, Euler's identity reads $J(x)x = 2(q_1(x), \dots, q_e(x))^t$. On charts where a source coordinate and a target coordinate are nonzero, choose the radial vectors first. The radial differential is a unit; its quotient is the projective differential. Removing that invertible block identifies the two ramification Fitting ideals. The contraction matrix is J^t , up to bases, so Lemma 2.1 applies.

A functional on S_R vanishing on $\gamma_R(HV)$ pulls back to one vanishing on $R + HV$. If $H = \ker \alpha$, then $(HV)^\perp = \mathbb{C}\alpha^2$. Thus such a nonzero functional exists exactly when all $q \in R$ vanish at $[\alpha]$. $\square$

Proposition 2.3 (The universal smooth open). *The set G_e° is nonempty, open, and $\mathrm{PGL}(V)$ -invariant. The determinant construction over it is a relative effective Cartier divisor $\mathcal{Y} \subset G_e^\circ \times P$, proper and smooth over G_e° . It is obtained by descent from the frame bundle of the universal quotient, and its formation commutes with arbitrary base change. Each fibre is integral, with trivial canonical bundle; for $e = 4$ it also has $H^1(\mathcal{O}) = 0$.*

Proof. Fix a p -dimensional space S and set $T = \mathrm{Hom}(\mathrm{Sym}^2 V, S)$. Restriction over $T \times P$ to $\mathrm{Sym}^2 \mathcal{H}$ is a surjective evaluation map to the rank- p^2 bundle of square matrices. The incidence of corank at least two has codimension four on each fibre over P , hence dimension $\dim T + e - 5 < \dim T$. Its projection to T is proper and has proper closed image. Similarly, HV has dimension

$p + e - 1$; nonsurjectivity of its restriction has codimension e and universal dimension $\dim T - 1$. Remove these two images and nonsurjective tensors. Also require invertibility of the square restriction at one fixed hyperplane. Each is a nonempty open condition on the irreducible space T , so their intersection T_1 is nonempty.

On $T_1 \times P$ the universal determinant section is nonzero on every fibre. Since a nonzero homogeneous polynomial is a non-zero-divisor on projective space over a field, its zero scheme Z is a relative Cartier divisor and is flat over T_1 . More explicitly, apply the fibrewise flatness criterion to $0 \rightarrow \mathcal{O}(-e) \rightarrow \mathcal{O} \rightarrow \mathcal{O}_Z \rightarrow 0$: the first two terms are base-flat and the first map stays injective on every fibre. This also proves that the same sequence stays exact after any base change.

The total space Z is smooth over $\mathbb{C}$. Indeed, evaluation is a submersion in the tensor directions, and the determinant hypersurface of square matrices is smooth at matrices of corank one; the corank-two incidence was removed. Every fibre is a nonempty degree- e hypersurface. The generic fibre over T_1 is regular, since its local rings are localizations of the regular local rings of Z . In characteristic zero it is geometrically regular, hence smooth. The nonsmooth locus of $Z \rightarrow T_1$ is closed, and its image is closed by properness and misses the generic point. Deleting that image gives a nonempty open with smooth fibres.

Surjections $\mathrm{Sym}^2 V \twoheadrightarrow S$ form the frame bundle of the rank- p universal quotient on $\mathrm{Gr}(e, \mathrm{Sym}^2 V)$. The conditions of basepoint-freeness and smooth determinant fibre depend only on the kernel. Their bad loci have closed proper projections from the corresponding projective incidences. They therefore define the asserted open downstairs. The determinant ideal, relative Cartier property, and flatness descend along the faithfully flat frame bundle; flatness with smooth geometric fibres gives smoothness. The cokernel and Fitting constructions themselves commute with arbitrary base change. The extra fixed-hyperplane condition above is used only to produce a nonempty open in T ; it is not imposed in the definition downstairs.

A smooth positive-degree hypersurface in $\mathbb{P}^{e-1}$ is integral here: distinct positive-degree components would meet and give a singular point, and repeated components are not smooth. Adjunction gives $\omega_{Y_R} = \mathcal{O}_{Y_R}$. The hypersurface sequence gives genus one when $e = 3$ and $H^1(Y_R, \mathcal{O}) = 0$ when $e = 4$. Thus the latter is a K3 surface. Smoothness of the determinant also excludes corank at least two in the square restriction, since all cofactors then vanish. $\square$

Lemma 2.4. *The algebras B_R and $B_{R'}$ are isomorphic exactly when R and R' lie in the same $\mathrm{PGL}(V)$ -orbit.*

Proof. An isomorphism preserves the maximal ideal and its square. On their associated graded spaces it induces an isomorphism of V carrying the kernel of quadratic multiplication to the other kernel. Conversely such a linear isomorphism gives an isomorphism of the graded algebras. Scalar matrices act trivially on relation subspaces. $\square$

3. THE MULTIPLICATION IDEAL AND INTRINSIC TORSION

We first calculate on the whole Grassmannian. In this section $R \in G_e^\circ$ is fixed, and the subscript R is omitted.

Proposition 3.1 (The full Fitting presentation). *On a frame chart write $\mathcal{K} \rightarrow V \oplus S$ as $\begin{pmatrix} M \\ A_0 \end{pmatrix}$. For every $m \geq 2$,*

$$\mathcal{I}_{\hat{D}} = (\det M)I_p(\gamma \mathrm{Sym}^2 M). \quad (3.1)$$

The reduction of $\hat{D}$ is the integral divisor $\hat{\Delta}$. Its smooth locus is the rank- $e - 1$ stratum. Consequently D is the restriction of $\hat{D}$ to the smooth locus of its reduction, and this restriction is intrinsic.

Proof. Cube-zero multiplication and the unit show, over the coefficient ring and not only on fibres, that the image of every degree $m \geq 2$ is $\mathbb{C}1 + \mathcal{K} + \mathcal{K}^2$. Its presentation is

$$\begin{pmatrix} 1 & 0 & 0 \\ 0 & M & 0 \\ 0 & A_0 & \gamma \operatorname{Sym}^2 M \end{pmatrix}.$$

Every nonzero maximal minor uses the scalar column and all e columns meeting the V -rows. Expanding on these rows proves (3.1). The graph block A_0 has no effect on this ideal. Off $\det M = 0$, the quadratic block is surjective because γ is surjective, so the ideal is the unit ideal. On the divisor it vanishes. The divisor is the usual irreducible Schubert divisor: the rank- $e - 1$ stratum is dense and integral, and every lower-rank plane is a specialization of that stratum. Equivalently the standard Schubert charts give its reduced determinantal equations.

Near a point with projection rank r , an invertible r -minor reduces the projection to $\operatorname{diag}(I_r, T)$, where the entries of the $(e - r) \times (e - r)$ block T are independent transverse coordinates. One can see their independence by varying the V -components of a basis of $K \cap S$ while keeping an invertible S -frame block fixed. The determinant equation is $\det T$. Its gradient is nonzero exactly at residual corank one. This proves the smooth-locus claim, including its assertion on the Grassmannian rather than on a chosen frame space. $\square$

On Δ put $H = \operatorname{im}(K \rightarrow V)$ and $\Lambda = K \cap S$. These give a morphism to $P \times \mathbb{P}(S)$. Define $E = \Delta \times_P Y$. The graph of K/Λ yields isomorphisms

$$\Delta \simeq \operatorname{Tot}_{P \times \mathbb{P}(S)} \operatorname{Hom}(\mathcal{H}, S/\Lambda), \quad (3.2)$$

$$E \simeq \operatorname{Tot}_{Y \times \mathbb{P}(S)} \operatorname{Hom}(\mathcal{H}, S/\Lambda). \quad (3.3)$$

Indeed the inverse construction takes the inverse image of a graph $H \rightarrow S/\Lambda$ in $H \oplus S$. Both constructions commute with base change. In particular E and Δ are smooth integral, of dimensions $ep - 2$ and $ep - 1$.

Theorem 3.2 (Ramification normal form). *On all of U the ideal of D is*

$$\mathcal{I}_D = \mathcal{I}_\Delta \mathcal{I}_E = \mathcal{I}_\Delta \cap \mathcal{I}_E^2. \quad (3.4)$$

The associated supports are exactly Δ and E . The nilradical is the invertible $\mathcal{O}_E$ -module $\mathcal{N} = \mathcal{O}_E(-\Delta)$ and is square-zero. Its annihilator recovers E as a scheme.

Proof. Near projection corank one, Gaussian operations make $M = \operatorname{diag}(I_{e-1}, t)$. The quadratic block becomes $[A, tB, t^2c]$, with $A|_{t=0} = \gamma|_{\operatorname{Sym}^2 H}$. Equation (2.1) says $[A, B]$ is onto locally; choose $AU + BV = I_p$. Multiplication by t then shows that $[A, tB, t^2c]$ and $[A, tI_p]$ have exactly the same column module. Their zeroth Fitting ideals agree. Off Y , A is invertible and (3.1) is (t) . On Y , the restriction has rank $p - 1$ by Proposition 2.3. A Schur complement gives a residual row with ideal (t, f) , where $f \bmod t$ is a local ramification equation. Thus the full ideal is

$$(t^2, tf) = t(t, f). \quad (3.5)$$

The pair t, f is a regular parameter pair cutting out E . The identity $(t) \cap (t, f)^2 = t(t, f)$ proves (3.4). These are primary ideals because their associated graded rings are polynomial rings over the integral smooth supports. At the generic point of E , neither primary component can be omitted. Their radicals are distinct, giving exactly the stated associated points. Finally the nilradical is $(t)/t(t, f)$; cancellation of t gives its module identification and annihilator $(t, f)/t(t, f)$. Both are intrinsic, so the calculation glues. $\square$

Lemma 3.3 (Generic torsion as a sheaf). *Let Z be noetherian with integral reduction and generic point η . For a coherent module $\mathcal{F}$ define*

$$\operatorname{tor}_\eta(\mathcal{F}) = \ker(\mathcal{F} \longrightarrow i_* \widetilde{\mathcal{F}}_\eta), \quad i : \operatorname{Spec} \mathcal{O}_{Z, \eta} \longrightarrow Z.$$

This kernel is coherent and is the largest coherent submodule with zero generic stalk. It is preserved by isomorphisms of schemes.

Proof. On an affine open meeting η this is the localization kernel at the prime of η . It is a submodule of a finite module over a noetherian ring, hence finite. Localization of that kernel on smaller affine opens is the corresponding kernel, so these modules glue. Every submodule with zero generic stalk lies in the kernel, and the kernel itself has zero generic stalk. The definition involves no ambient embedding or chosen primary decomposition. $\square$

Corollary 3.4 (Ordinary powers). *For $n \geq 1$, write $D^{[n]} = V(\mathcal{I}_D^n)$ and $E^{[n]} = V(\mathcal{I}_E^n)$. Then*

$$\mathcal{I}_D^n = \mathcal{I}_\Delta^n \mathcal{I}_E^n = \mathcal{I}_\Delta^n \cap \mathcal{I}_E^{2n} \quad (3.6)$$

is an irredundant primary decomposition on U . With $\mathcal{T}_n = \text{tor}_{\eta_\Delta}(\mathcal{O}_{D^{[n]}})$,

$$0 \rightarrow \mathcal{T}_n = \mathcal{O}_{E^{[n]}}(-n\Delta) \rightarrow \mathcal{O}_{D^{[n]}} \rightarrow \mathcal{O}_{n\Delta} \rightarrow 0. \quad (3.7)$$

Its annihilator defines $E^{[n]}$. The generic multiplicity on Δ is n , the nilradical index along E is $2n$, and the length of $\mathcal{T}_n$ at η_E is $n(n+1)/2$.

Proof. In the local ring let $P = (t, f)$. The associated graded ring of P is a polynomial ring in two variables over $\mathcal{O}_E$, and the initial form of t is one of those variables. Hence $(P^{2n} : t^n) = P^n$, proving $(t^n) \cap P^{2n} = t^n P^n$. The same graded argument proves that these powers are primary. The witnesses t^n and f^{2n} at η_E prove irredundancy. Localizing at (t) contracts the ideal to (t^n) , so the generic torsion is $(t^n)/t^n P^n$, and cancellation gives its annihilator P^n . The class of t has index $2n$ at E , since t^{2n} belongs to the ideal but t^{2n-1} does not. At the generic point of Δ , f is a unit. Finally $\mathcal{O}_{U, \eta_E}/(t, f)^n$ has the $n(n+1)/2$ monomials of total degree less than n as successive graded basis elements. $\square$

Lemma 3.5 (Flat flags and arbitrary base change). *Let $U \rightarrow T$ be smooth, and let $E \subset \Delta \subset U$ be relative smooth closed subschemes, each inclusion a Cartier divisor. Put $L = \mathcal{I}_\Delta$ and $P = \mathcal{I}_E$. Then $\mathcal{O}_U/P^n$ and $\mathcal{O}_U/L^n$ are flat over T . For $D_n = V(L^n P^n)$ the sequence*

$$0 \rightarrow L^n \otimes \mathcal{O}_{E^{[n]}} \rightarrow \mathcal{O}_{D_n} \rightarrow \mathcal{O}_U/L^n \rightarrow 0 \quad (3.8)$$

is exact and remains exact after any base change. Its base-changed terms are the indicated ordinary powers and products. In particular D_n is flat over T .

Proof. The immersion $E \subset U$ is regular of codimension two and has locally free conormal module. The conormal filtration of $\mathcal{O}_U/P^n$ has quotients $\text{Sym}^j(P/P^2)$ on E , for $0 \leq j < n$; these are base-flat. Induction proves flatness. The analogous principal filtration proves flatness of $\mathcal{O}_U/L^n$. Multiplication by a local equation t^n of L^n identifies the kernel in (3.8) with $L^n \otimes \mathcal{O}_{E^{[n]}}$, since t is a non-zero-divisor. Both outer terms are base-flat, so the middle term is base-flat. For a base algebra $A \rightarrow A'$, the only possible left obstruction in tensoring this short exact sequence is $\text{Tor}_1^A(\mathcal{O}_U/L^n, A')$, which is zero. The sequences defining L , P , and their powers stay exact for the same reason. Thus their images after base change are the actual ideals generated by the base-changed equations, and multiplication gives their stated products. No assertion about specialization of selected primary components is used. $\square$

Corollary 3.6. *Over G_e° , the schemes E , Δ , and U form the smooth relative flag of Lemma 3.5. All $D^{[n]}$ and $E^{[n]}$ are flat, and (3.7) commutes with arbitrary base change. Each complex fibre has (3.6).*

Proof. Use the universal quotient $\mathcal{S}$ and the graph bundles (3.2)–(3.3) over the smooth family of Proposition 2.3. The Fitting elimination works over the parameter ring, so it gives $t(t, f)$ relatively. Apply the lemma, and apply Corollary 3.4 separately on each fibre for the primary assertion. $\square$

4. THE INTRINSIC NILPOTENT-DEPTH FILTRATION

We isolate the functorial mechanism that unifies the two local models. Let Z be any noetherian scheme and let $\mathcal{N}_Z$ denote the nilradical of $\mathcal{O}_Z$. Since Z is noetherian, $\mathcal{N}_Z$ is a coherent nilpotent ideal. For $j \geq 1$ define

$$\mathcal{Z}_j(Z) = V\left(\mathrm{Ann}_{\mathcal{O}_Z}(\mathcal{N}_Z^j)\right). \quad (4.1)$$

Lemma 4.1 (Functoriality). *Every isomorphism $f : Z \rightarrow Z'$ carries $\mathcal{Z}_j(Z)$ isomorphically to $\mathcal{Z}_j(Z')$ for every j . The construction commutes with restriction to open subsets.*

Proof. An isomorphism carries the nilradical to the nilradical, hence carries its j -th power and its annihilator ideal to the corresponding objects. Both formation of powers and annihilators commute with localization. $\square$

Apply this to $Z = \widehat{D}_R$. On the intrinsic open obtained by restricting to the smooth locus of Z_{red} , Proposition 3.1 identifies Z with D_R . Theorem 3.2 gives the local ring

$$A = \mathcal{O}/(t(t, f))$$

with nilradical generated by the class of t . Therefore

$$\mathrm{Ann}_A((t)) = (t, f)/(t(t, f)),$$

so (4.1) recovers E_R scheme-theoretically.

Proposition 4.2 (Depth one is the polarized ramification support). *On the smooth-reduction locus of $\widehat{D}_R$,*

$$\mathcal{Z}_1(\widehat{D}_R) = E_R, \quad \widehat{\mathcal{N}}_R|_{E_R} = \mathcal{O}_{E_R}(-\widehat{\Delta}_R).$$

Consequently the pair

$$\left(\mathcal{Z}_1(\widehat{D}_R), \omega_{\mathcal{Z}_1(\widehat{D}_R)} \otimes (\widehat{\mathcal{N}}_R|_{\mathcal{Z}_1(\widehat{D}_R)})^{-9}\right)$$

is intrinsic when $e = 4$, and its graded section ring is the fourth Veronese of the polarized quartic section ring.

Proof. Only the last assertion is new notation. The annihilator calculation above is Theorem 3.2; the line-bundle identity and section-ring reconstruction are Proposition 5.2 and Theorem 1.1. $\square$

The point of (4.1) is that it makes no reference to a projection chart. The closed schemes are automatically nested:

$$\widehat{\mathcal{N}}_R^{j+1} \subset \widehat{\mathcal{N}}_R^j \implies \mathrm{Ann}(\widehat{\mathcal{N}}_R^j) \subset \mathrm{Ann}(\widehat{\mathcal{N}}_R^{j+1}), \quad \mathcal{Z}_{j+1} \subset \mathcal{Z}_j.$$

The new issue is therefore not formal functoriality but the scheme-theoretic computation of the layers as one crosses projection rank. Theorem 9.1 solves this globally: on the whole Grassmannian the residual colon

$$\mathcal{J}_R^{\mathrm{res}} = (\mathcal{I}_{\widehat{D}_R} : \mathcal{I}_{\widehat{\Delta}_R})$$

determines every annihilator and every associated-graded piece, including all projection coranks and rank-drop loci. The remaining task is geometric identification of those canonical colon schemes.

On the first higher-corank chart the next power is nonzero. The calculation in §7 gives, on the simple-contact open,

$$\widehat{\mathcal{N}}_R^3 = 0, \quad \mathrm{Ann}(\widehat{\mathcal{N}}_R^2) = \mathcal{I}_{\Sigma_R}/\mathcal{I}_{\widehat{D}_R}.$$

Section 8 removes the simple-root assumption. Section 9 computes the two layers simultaneously across the corank-one/corank-two boundary. On the mixed-regular neighbourhood,

$$\mathrm{gr}_{\widehat{\mathcal{N}}_R} \mathcal{O}_{\widehat{D}_R} = \mathcal{O}_{\widehat{\Delta}_R} \oplus (\mathcal{L} \otimes \mathcal{O}_{C_R}) \oplus (\mathcal{L}^{\otimes 2} \otimes \mathcal{O}_{\Sigma_R}),$$

where $\mathcal{C}_R$ is the unsplit contact cone and $\mathcal{L}$ is the Schubert conormal line. In particular $\mathcal{Z}_1$ specializes from the ramification support to its canonical contact closure with the vertex thickening forced by the Fitting algebra, while $\mathcal{Z}_2$ is the rank-two vertex. On the first tensor-rank boundary of the mixed map, $\mathcal{Z}_2$ acquires a canonical length-three nonreduced structure (Theorem 9.5).

Thus

$$\mathcal{Z}_1 \rightsquigarrow \mathcal{Z}_2$$

is not merely a comparison of two calculations on disjoint strata: it is the associated-graded specialization of one intrinsic nilpotent algebra.

Remark 4.3 (Relation with canonical filtrations on multiple structures). Nilpotent and multiple structures have long been studied through canonical filtrations and associated graded modules; ribbons are the multiplicity-two prototype, and primitive multiple schemes carry a canonical chain of lower multiplicities. See [4, 15, 11]. The construction (4.1) is deliberately more elementary: it is available for the present Fitting schemes without assuming that they are primitive, locally complete-intersection multiple structures, or Cohen–Macaulay along every higher-corank stratum. The additional content here is that consecutive annihilator layers and their associated graded modules have a specific geometric meaning forced by multiplication: the first carries the polarized ramification K3, the second detects the rank-two Schubert support, and their specialization through contact collisions is controlled by the unsplit contact Cartier ideal rather than by a choice of primary roots.

5. RECOVERING THE POLARIZED RAMIFICATION VARIETY

Let $A = \mathcal{O}_Y(1)$ and $B = \mathcal{O}_{\mathbb{P}(S)}(1)$. In the graph description (3.3) write

$$Z = Y \times \mathbb{P}(S), \quad W = \mathcal{H}^* \otimes Q, \quad Q = (S \otimes \mathcal{O})/\Lambda, \quad \pi : E = \text{Tot}_Z W \rightarrow Z,$$

and let $\rho : E \rightarrow Y$ be the composite projection. Constant one-dimensional vector-space factors in determinant lines below are suppressed; they change none of the isomorphism classes or section rings.

Lemma 5.1 (No extra fibrewise functions). *The natural map $\mathcal{O}_Y \rightarrow \rho_* \mathcal{O}_E$ is an isomorphism.*

Proof. The assertion is local on Y , where $\mathcal{H}$ can be trivialized. If H_0 is its fibre space, the affine bundle projection gives

$$\pi_* \mathcal{O}_E = \bigoplus_{d \geq 0} \text{Sym}^d(H_0 \otimes Q^*).$$

We claim that

$$H^0(\mathbb{P}(S), \text{Sym}^d(H_0 \otimes Q^*)) = 0 \quad (d > 0). \quad (5.1)$$

The injection $Q^* \hookrightarrow S^* \otimes \mathcal{O}$ embeds a global section in the constant space $\text{Sym}^d(H_0 \otimes S^*)$. Such an element is a homogeneous polynomial on $\text{Hom}(H_0, S)$. At every line $\Lambda \subset S$, membership in $\text{Sym}^d(H_0 \otimes (S/\Lambda)^*)$ says this polynomial factors through $\text{Hom}(H_0, S/\Lambda)$; it is invariant under translations by $\text{Hom}(H_0, \Lambda)$. Taking the lines of a basis of S shows invariance under every translation of $\text{Hom}(H_0, S)$. In characteristic zero such a polynomial is constant, so its positive-degree part is zero. This proves (5.1). The degree-zero part is $\mathbb{C}$.

The same calculation tensors with the coordinate ring of each affine trivializing open of Y : cohomology on the projective space commutes with this flat extension from $\mathbb{C}$, and with the displayed direct sum in degree zero. Hence the pushforward is precisely $\mathcal{O}_Y$, with its natural multiplication. The local identifications glue. $\square$

Proposition 5.2 (The canonical nilradical line). *As line bundles on E ,*

$$\mathcal{N} \simeq \pi^*(A^{-1} \boxtimes B^{-1}), \quad \omega_E \simeq \pi^*(A^{-(p-1)} \boxtimes B^{-(p+e-1)}). \quad (5.2)$$

Consequently

$$\omega_E \otimes \mathcal{N}^{\otimes -(p+e-1)} \simeq \rho^* A^e. \quad (5.3)$$

Proof. The determinant defining $\hat{\Delta}$ is a section of $(\det \mathcal{K})^* \otimes \det V$. On E the exact sequence $0 \rightarrow \Lambda \rightarrow \mathcal{K} \rightarrow \mathcal{H} \rightarrow 0$ gives $\mathcal{I}_{\Delta}|_E = \det \mathcal{H} \otimes \Lambda$, up to a constant factor. Since $\det \mathcal{H} = A^{-1}$ and $\Lambda = B^{-1}$, Theorem 3.2 gives the first formula.

The determinant of W is $(\det \mathcal{H}^*)^{p-1} \otimes (\det Q)^{e-1} = A^{p-1} \boxtimes B^{e-1}$. The canonical bundle of a vector-bundle total space is $\pi^*(\omega_Z \otimes \det W^{-1})$, as follows by taking the determinant of its relative cotangent sequence. Here $\omega_Z = \mathcal{O}_Y \boxtimes B^{-p}$ because $\omega_Y = \mathcal{O}_Y$. This proves the second formula. Subtracting the two exponents gives (5.3): the B -exponent vanishes and the A -exponent is $-(p-1) + (p+e-1) = e$. $\square$

Proof of Theorem 1.1. An isomorphism of failure schemes preserves the nilradical and its annihilator, hence E , its invertible module $\mathcal{N}$, and its canonical bundle. It therefore preserves $\mathcal{L} = \omega_E \otimes \mathcal{N}^{-(p+e-1)}$ and its graded section ring. Proposition 5.2, Lemma 5.1, and the projection formula give

$$H^0(E, \mathcal{L}^n) = H^0(Y, A^{en} \otimes \rho_* \mathcal{O}_E) = H^0(Y, A^{en})$$

compatibly with multiplication. The full section ring of the ample line bundle A^e recovers (Y, A^e) by Proj. In this case it is also the e th Veronese of the homogeneous coordinate ring of a smooth hypersurface, so there is no ambiguity about the grading line bundle.

For a K3 surface the Picard group has no torsion. Here is the short argument needed: a nontrivial torsion line bundle T and its inverse have no nonzero sections, since a nonzero effective torsion divisor has zero intersection with an ample divisor, which is impossible. Riemann–Roch and Serre duality would then give $2 = \chi(T) = -h^1(T) \leq 0$, a contradiction. Thus an isomorphism carrying A^4 to $(A')^4$ carries A to A' . This recovers the quartic polarization, and its complete linear system recovers the quartic embedding up to projective equivalence.

Finally Proposition 3.1 recovers D from $\hat{D}$ by restricting to the smooth locus of the reduction. This operation is intrinsic and is preserved by every abstract scheme isomorphism. The preceding proof applies to that restriction. $\square$

Remark 5.3. For $e = 3$ the theorem recovers (Y, A^3) directly. A chosen cubic root in the Picard group need not be unique. As polarized genus-one curves, however, the possible degree-three roots are isomorphic: after choosing an origin, the map $y \mapsto t_y^* A \otimes A^{-1}$ onto $\text{Pic}^0(Y)$ is multiplication by three, hence surjective. No choice of an origin is needed for the stated section-ring reconstruction. For $e = 4$ the torsion-free argument gives uniqueness without translations.

Remark 5.4 (Relation to the earlier birational argument). The compactification $\mathbb{P}_Z(\mathcal{O} \oplus W)$ is birational to $Y \times \mathbb{P}^{e(p-1)}$. Thus the earlier cancellation proof is a special case of the maximal rationally connected quotient: for a non-uniruled base, that quotient of a product with projective space is the base, birationally. See the rational-quotient construction in [14, Chapter IV, §5]. This explains the earlier abstract recovery, which is retained in the companion. The present proof does not rely on that argument: the section ring gives the actual projective model and its line bundle, not merely a birational class.

6. NINE-DIMENSIONAL VARIATION IN THE K3 CASE

We prove Theorem 1.4. Work on the standard chart of $\text{Gr}(4, \text{Sym}^2 \mathbb{C}^4)$ where the four square coefficients are the identity. Write $x = (x_0, x_1, x_2, x_3)$ and order the six cross terms as

$$m = (x_0 x_1, x_0 x_2, x_0 x_3, x_1 x_2, x_1 x_3, x_2 x_3).$$

For $C \in \text{Mat}_{6 \times 4}$ set

$$q_i = x_i^2 + \sum_{k=1}^6 C_{ki} m_k, \quad J_C = (\partial q_i / \partial x_j), \quad F_C = \det J_C. \quad (6.1)$$

This parametrizes the relation space, not an additional choice of its basis: the square coefficients fix that basis on this chart.

Lemma 6.1 (An integral differential witness). *At the relation system*

$$C_* = \begin{pmatrix} 2 & -1 & 2 & 0 \\ -2 & 0 & -1 & 1 \\ 2 & -1 & -2 & 0 \\ 1 & -1 & 2 & 1 \\ 1 & -1 & 0 & 0 \\ 1 & -2 & 1 & -2 \end{pmatrix}, \quad (6.2)$$

the differential of $C \mapsto [F_C] \in \mathbb{P}(\text{Sym}^4 \mathbb{C}^4)$ has rank 24. This relation system is basepoint-free and $V(F_{C_*})$ is smooth over $\mathbb{C}$.

Proof. Let $\text{cof}_{ij}(J)$ denote the signed cofactor. The 24 parameter derivatives are

$$G_{ik} = \left. \frac{\partial F_C}{\partial C_{ki}} \right|_{C_*} = \sum_{j=0}^3 \text{cof}_{ij}(J_{C_*}) \frac{\partial m_k}{\partial x_j}. \quad (6.3)$$

Order degree-four monomials by descending lexicographic order of their exponent tuples, and form the 35×25 integral coefficient matrix D_* with columns F_{C_*} , followed by G_{ik} with $i = 0, \dots, 3$ outermost and $k = 1, \dots, 6$ innermost. The determinant of its first 25 rows and all columns is

$$4279473148893659522379284480 \neq 0. \quad (6.4)$$

Its residue modulo 101 is 41. Thus F_{C_*} and the 24 derivatives are linearly independent. Quotienting the target tangent space by the line spanned by F_{C_*} proves the differential assertion.

For completeness, Appendix B specifies the full coefficient recipe and two additional finite rank certificates. The Macaulay matrix of the q_i in degree five has rank 56, the dimension of all degree-five polynomials. The analogous matrix of the four partials of F_{C_*} in degree nine has rank 220, again the entire homogeneous piece. Explicit square minors are nonzero modulo 101, with residues 44 and 79. The integer matrices therefore have full row rank over $\mathbb{Q}$. Every monomial of those degrees belongs to the respective homogeneous ideal over $\mathbb{Q}$, and hence over $\mathbb{C}$. A common projective zero is impossible. This proves basepoint-freeness and smoothness without a numerical tolerance or a probabilistic test. $\square$

Proposition 6.2. *The Jacobian map on G_4° has a 24-dimensional image in the space of embedded smooth quartics. Its image in polarized quartic K3 moduli has dimension nine. Its general fibres after projective coordinate equivalence are finite.*

Proof. Changing a basis of R multiplies the determinant by a nonzero scalar, so the projective Jacobian map is well-defined. A change of coordinates in V acts equivariantly. Lemma 6.1 and the 24-dimensional domain show that its image has dimension 24 and that the map is generically finite onto that embedded image. The locus where the differential is injective is nonempty and open; it is invariant because the map is equivariant. The smooth admissible locus is irreducible by Proposition 2.3.

The projective stabilizer of a smooth quartic K3 is finite. Indeed, contraction with its nonzero two-form identifies its tangent bundle with its cotangent bundle, and $H^0(\Omega^1) = 0$ by $H^1(\mathcal{O}) = 0$ and Hodge symmetry over $\mathbb{C}$. The stabilizer, a closed finite-type subgroup of PGL_4 , therefore

has zero-dimensional Lie algebra and is finite. Its orbits have dimension 15. Two smooth quartic embeddings define the same polarized K3 precisely when they are related by PGL_4 , because their embeddings are given by the complete polarizing linear systems. The geometric quotient of the smooth quartic locus gives the quartic K3 moduli space; equivalently one may compute on a generic geometric quotient. The invariant image consequently has dimension $24 - 15 = 9$ after quotienting. Standard background on polarized K3 moduli is given in [12, Chapter 5].

To check the finiteness assertion directly, take a general embedded quartic in the 24-dimensional image where the fibre is finite. Every relation system mapping to a projectively equivalent quartic can be moved by the corresponding projective transformation into that finite fibre. Thus there are only finitely many projective equivalence classes above the general polarized point. This argument also makes clear that the assertion is generic, rather than a statement about all exceptional fibres. $\square$

Proof of Theorem 1.4. Combine Proposition 6.2 with the recovery of the polarized quartic in Theorem 1.1 and the algebra-orbit equivalence of Lemma 2.4. $\square$

Corollary 6.3. *There is a flat family of failure schemes on the corank-one open whose canonical polarized invariant has a nine-dimensional image. In particular the associated K3 surfaces are not isotrivial, and the failure schemes are not all abstractly isomorphic. Their reduced divisors become the same scheme after identifying the vector spaces S_R of rank six.*

Proof. Flatness is Corollary 3.6. A fixed abstract K3 cannot give a positive-dimensional algebraic family of polarization classes on that surface: its Picard scheme is discrete, since $H^1(\mathcal{O}) = 0$. Thus the positive-dimensional polarized image also implies non-isotriviality of the surfaces. Distinct values of the canonical polarized invariant give nonisomorphic failure schemes by Theorem 1.1. The graph description of Δ in (3.2) is independent of R . $\square$

7. FIXED TENSORS ACROSS PROJECTION CORANK TWO

We now work with $e = 4$ and $p = 6$. All calculations below concern a fixed quotient tensor γ_R , while its generating plane varies. They are identities of ideals, before taking radicals.

Definition 7.1. A pair (R, H) , where $R \in G_4^\circ$ and $H \subset V$ is a plane, is called admissible for the corank-two calculation if the following conditions hold. The map $A_H = \gamma_R|_{\mathrm{Sym}^2 H}$ is injective. For a complement W to H , put $Q_H = S_R / \mathrm{im} A_H$ and let

$$B_H : H \otimes W \longrightarrow Q_H, \quad C_H : \mathrm{Sym}^2 W \longrightarrow Q_H$$

be induced by multiplication. The map B_H is surjective and its one-dimensional kernel is generated by a rank-two tensor in $H \otimes W$. Finally, the line of hyperplanes containing H meets Y_R in four distinct points.

The first-order map B_H is intrinsically a map from $H \otimes (V/H)$: changing a lift of a vector in V/H adds an element of $\mathrm{Sym}^2 H$, which is zero in Q_H . Thus these conditions are independent of the chosen complement. The last condition is transverse intersection with a projective line, not smoothness of a parameter-selected polynomial without a geometric interpretation.

Lemma 7.2 (Nonemptiness of the corank-two open). *The admissible pairs form a nonempty open subset of $G_4^\circ \times \mathrm{Gr}(2, V)$. Its projection is a nonempty invariant open $G_4^\dagger$. For each R in that image, the admissible H form a dense open subset of $\mathrm{Gr}(2, V)$.*

Proof. Injectivity and surjectivity are open rank conditions. After taking the line kernel, tensor rank two is the nonvanishing of its determinant. On this locus the pencil restriction of the determinant is a binary quartic; nonzero discriminant is open. These conditions are invariant.

Projection from a product over $\mathbb{C}$ is open, and $\text{Gr}(2, V)$ is irreducible, giving the density claim once nonemptiness is proved.

For C_* in (6.2), take $H = \langle x_0, x_1 \rangle$ and $W = \langle x_2, x_3 \rangle$. In the square-first, cross-second basis, $\gamma = (-C_* \mid I_6)$. The three columns for x_0^2, x_0x_1, x_1^2 have rank three. After quotienting their image, the remaining maps are

$$B_H = \begin{pmatrix} -1/2 & -1 & 1 & 0 \\ -1/2 & -1 & 0 & 1 \\ -3/2 & -2 & 0 & 0 \end{pmatrix}, \quad C_H = \begin{pmatrix} -9/2 & 0 & -1/2 \\ -5/2 & 0 & 1/2 \\ -13/2 & 1 & 7/2 \end{pmatrix}. \quad (7.1)$$

The four columns of B_H correspond to $x_0x_2, x_0x_3, x_1x_2, x_1x_3$; the three of C_H correspond to x_2^2, x_2x_3, x_3^2 . The kernel vector is $(-4, 3, 1, 1)$, whose 2×2 determinant is -7 . For $ux_2 + vx_3$, the final square column is $C_H(u^2, 2uv, v^2)^t$. The determinant of that column and the two mixed columns is

$$h(u, v) = \frac{u^4 - 8u^3v - u^2v^2 - 4uv^3 - 9v^4}{2}, \quad \text{disc } h(u, 1) = -\frac{632301}{4} \neq 0. \quad (7.2)$$

Its leading coefficient is nonzero, so there is no root at infinity. This determinant is the restriction determinant on $\text{Sym}^2(H + \mathbb{C}(ux_2 + vx_3))$ after eliminating A_H , hence defines the pencil intersection with Y_R . Lemma 6.1 places this witness in G_4° . $\square$

Lemma 7.3 (The residual matrix). *Near a plane K whose projection has admissible rank-two image H , there are smooth auxiliary parameters and four transverse coordinates a, b, c, d in which $\mathcal{I}_{\widehat{D}_R} = \delta J$, with*

$$\begin{aligned} T &= \begin{pmatrix} a & b \\ c & d \end{pmatrix}, \quad \delta = ad - bc, \quad \mathbf{m} = (a, b, c, d), \\ L(T) &= \begin{pmatrix} a & b & 0 & 0 \\ c & d & a & b \\ 0 & 0 & c & d \end{pmatrix}, \quad T^{(2)} = \begin{pmatrix} a^2 & ab & b^2 \\ 2ac & ad + bc & 2bd \\ c^2 & cd & d^2 \end{pmatrix}, \\ J &= I_3[L(T), C T^{(2)}]. \end{aligned} \quad (7.3)$$

Here C is a 3×3 matrix depending only on the auxiliary parameters, not on a, b, c, d .

Proof. Use Proposition 3.1 and an invertible projection two-minor to write $M = \text{diag}(I_2, T)$. The basis of its first image plane and a complement can be chosen independently of the Schur block T : the Gaussian row operations depend on the first two columns, and the remaining column operations remove their components from the last two columns. This proves the asserted separation of parameters.

The quadratic columns divide into $\text{Sym}^2 H$, $H \otimes W$, and $\text{Sym}^2 W$. Eliminate the injective rank-three A_H block. The residual matrix is $[B_H(I_H \otimes T), C_H T^{(2)}]$, and its maximal minors generate the residual ideal: the eliminated identity block removes a free direct summand of the cokernel. A rank-two generator of $\ker B_H$ can be put in the form $h_1 \otimes w_2 - h_2 \otimes w_1$ by invertible changes of bases. The quotient is then the symmetrization map from two two-dimensional spaces, with the displayed matrix $L(T)$. These basis changes involve only the auxiliary parameters. Finally $\det M$ is a unit times δ . This proves the ideal identity, not just its leading term. $\square$

Let

$$h(u_1, u_2) = \det \left[\begin{pmatrix} u_1 \\ u_2 \\ 0 \end{pmatrix}, \begin{pmatrix} 0 \\ u_1 \\ u_2 \end{pmatrix}, C \begin{pmatrix} u_1^2 \\ 2u_1u_2 \\ u_2^2 \end{pmatrix} \right]. \quad (7.4)$$

It is the binary quartic of the pencil intersection. After an étale extension of the auxiliary coefficient ring, its four roots split as distinct line subbundles $\ell_i \subset W$. Write P_i for the ideal of $\text{int } T \subset \ell_i$; it is generated by the two entries of the composite $\mathbb{C}^2 \xrightarrow{T} W \rightarrow W/\ell_i$.

Lemma 7.4 (A rank-one-cone computation). *Under the simple-root hypothesis, in the local coefficient ring of Lemma 7.3 one has*

$$I := \bigcap_{i=1}^4 P_i = (\delta, g_0, \dots, g_4), \quad J = I \cap \mathfrak{m}^3 = \delta \mathfrak{m} + (g_0, \dots, g_4). \quad (7.5)$$

The degree-four polynomials g_j can be chosen with restrictions $h(u_1, u_2)v_1^{4-j}v_2^j$ on $T = uv^t$.

Proof. Give a, b, c, d degree one and the auxiliary coefficients degree zero. The coordinate ring of the rank-one cone is the diagonal subring

$$A_0[a, b, c, d]/(\delta) \simeq \bigoplus_{r \geq 0} \text{Sym}^r(A_0^2)_u \otimes \text{Sym}^r(A_0^2)_v, \quad T = uv^t, \quad (7.6)$$

where A_0 denotes the auxiliary ring. This follows directly from $a = u_1v_1$, $b = u_1v_2$, $c = u_2v_1$, $d = u_2v_2$ and the single relation $ad - bc$; the monomials on either side give the same basis. In degree r , vanishing on the four distinct u -lines means divisibility by h in the u variables. The resulting module is zero for $r < 4$, and for $r \geq 4$ it is

$$h \text{Sym}^{r-4}(A_0^2)_u \otimes \text{Sym}^r(A_0^2)_v.$$

The simple-root discriminant is a unit, so the elementary divisibility and interpolation assertions used here hold over A_0 as well as over its residue field. Multiplication of binary forms shows that this ideal is generated in diagonal degree four by $hv_1^{4-j}v_2^j$. Lifting those generators proves the first identity.

The four 3-minors of the linear block $L(T)$ generate $\delta \mathfrak{m}$, by direct expansion. Modulo δ , its four columns are

$$v_1B(h_1u), \quad v_2B(h_1u), \quad v_1B(h_2u), \quad v_2B(h_2u),$$

and the three quadratic columns are $v_1^2C(u^2)$, $v_1v_2C(u^2)$, and $v_2^2C(u^2)$. Here u^2 has coefficient vector $(u_1^2, 2u_1u_2, u_2^2)^t$. A minor using one column from each of the two linear groups and one quadratic column is h times a degree-four monomial in v ; these span all five such monomials. Thus they furnish the lifts g_j . A minor with at least two quadratic columns vanishes modulo δ , and its degree is at least five, so it already lies in $\delta \mathfrak{m}$. All other minors have been accounted for. This proves $J = \delta \mathfrak{m} + (g_j)$ exactly.

Finally I is generated by its degree-two element δ and the g_j of degree four. Its intersection with $\mathfrak{m}^3$ deletes only the scalar multiple of δ in degree two, proving the second equality. Localizing in the auxiliary parameters or taking the indicated étale extension preserves these explicit identities. $\square$

Theorem 7.5 (The corank-two primary structure). *For an admissible pair the ideal δJ has the decomposition (1.10). It is irredundant. In the local failure ring its nilradical $\widehat{\mathcal{N}}$ satisfies*

$$\widehat{\mathcal{N}}^3 = 0, \quad \widehat{\mathcal{N}}^2 \neq 0, \quad \text{Ann}(\widehat{\mathcal{N}}^2) = \mathfrak{m}/(\delta J). \quad (7.7)$$

Consequently the rank-two stratum is intrinsic to this local scheme as the annihilator support of $\widehat{\mathcal{N}}^2$.

Proof. Work on a connected regular étale auxiliary chart, so that its coefficient ring is a domain. Each P_i is a prime generated by two independent linear coordinates. Its ordinary square is P_i -primary; the same argument applies to $\mathfrak{m}^5$. The ideal (δ) is prime. At the generic point of P_i ,

the determinant has order one, because the other row after choosing ℓ_i as a coordinate line is not zero. Therefore

$$(P_i^2 : \delta) = P_i, \quad (\mathfrak{m}^5 : \delta) = \mathfrak{m}^3.$$

The second equality also follows immediately from the homogeneous order two of δ and the polynomial associated graded ring of $\mathfrak{m}$. Intersecting with (δ) and cancelling δ now gives

$$(\delta) \cap \bigcap_i P_i^2 \cap \mathfrak{m}^5 = \delta \left(\bigcap_i P_i \cap \mathfrak{m}^3 \right) = \delta J$$

by Lemma 7.4.

At the generic point of the determinant divisor all the other ideals are units, so (δ) is necessary. At the generic point of P_i the other P_j and $\mathfrak{m}$ become units, and the local ideal is $\delta P_i = (\delta) \cap P_i^2$; its two components are necessary. To see that $\mathfrak{m}^5$ is necessary, note that δ^2 lies in (δ) and in every P_i^2 , but is not in δJ : $\delta \notin J$ because its degree is two. On the other hand $\mathfrak{m}\delta^2 \subset \delta J$ since $\delta\mathfrak{m} \subset J$. This also proves that the annihilator of the class of δ^2 is exactly $\mathfrak{m}$; a nonzero auxiliary coefficient cannot kill that degree-four class.

The radical of the failure ideal is (δ) , so the nilradical is generated by the class of δ . Its square is the nonzero class just considered and its cube vanishes, since $\delta^2 \in \delta\mathfrak{m} \subset J$. This proves (7.7). Over the auxiliary ring the square is free of rank one on the rank-two stratum. Intrinsically its transition functions are those of $\mathcal{I}_{\hat{\Delta}}^2$ restricted to that stratum; thus it is $\mathcal{O}_{\Sigma}(-2\hat{\Delta})$ on the admissible open Σ . The generic transverse length is one. $\square$

Proof of Theorem 1.3. Lemma 7.2 supplies the asserted nonempty open class. Lemmas 7.3 and 7.4 and Theorem 7.5 supply the fixed-tensor calculation on every indicated frame chart. The graph coordinates are smooth auxiliary variables, so no transverse component has been discarded. The individual P_i are allowed to permute under monodromy; the intersection and the intrinsic annihilator in (7.7) descend. Four local branches need not be four distinct global irreducible components. This is why the primary decomposition was stated étale locally rather than with globally labelled roots. $\square$

8. RESIDUAL PRESENTATIONS AND THE RANK-TWO CONTACT BOUNDARY

We first record the projection-rank block presentation inherited from the full Fitting formula, and then isolate the rank-two structure to which it gives rise. The distinction between a presentation theorem and a structure theorem is important: the block formula below is exact at every projection rank, whereas the associated-prime, collision, and nilpotent-filtration theorems are proved on the loci explicitly stated. Section 9 pushes the structure theorem through all mixed-regular contact collisions, through the containment locus, and across the first non-mixed-regular tensor-rank wall.

For $0 \leq r \leq 3$, let $\Sigma_R^{(r)} \subset X_R$ be the locally closed stratum on which the projection $\mathcal{K} \rightarrow V$ has rank exactly r . If $K \in \Sigma_R^{(r)}$ has image $H \subset V$, choose a complement $V = H \oplus W$, where $c = \dim W = 4 - r$. On a frame chart around K , an invertible r -minor and Schur reduction put the projection block in the form

$$M = \begin{pmatrix} I_H & 0 \\ 0 & T \end{pmatrix}, \quad T \in \text{Mat}_{c \times c}.$$

Decompose the quotient multiplication tensor according to

$$\text{Sym}^2 V = \text{Sym}^2 H \oplus (H \otimes W) \oplus \text{Sym}^2 W$$

and write the corresponding blocks as A_H, B_H, C_H .

Theorem 8.1 (Projection-rank residual block presentation). *For every $R \in G_4^2$, every $0 \leq r \leq 3$, and every $K \in \Sigma_R^{(r)}$, one has on the preceding frame chart, before taking radicals,*

$$\mathcal{I}_{\widehat{D}_R} = (\det T) I_6[A_H, B_H(1_H \otimes T), C_H \operatorname{Sym}^2 T]. \quad (8.1)$$

The formula is independent of any injectivity, mixed-rank, tensor-rank, or contact-transversality assumption. Thus every projection-rank stratum has the same exact block presentation. No claim of a primary or associated-prime classification on the non-mixed-regular corank-three or corank-four boundary is made by this formula alone; the separate structure theorems below derive additional information where the residual algebra can be determined.

Proof. Proposition 3.1 gives on every frame chart

$$\mathcal{I}_{\widehat{D}_R} = (\det M) I_6(\gamma_R \operatorname{Sym}^2 M).$$

The Schur operations placing M in the displayed block form are invertible and therefore change neither the maximal-minor ideal nor the local scheme. On $\operatorname{Sym}^2 H \oplus (H \otimes W) \oplus \operatorname{Sym}^2 W$ the induced map $\operatorname{Sym}^2 M$ is respectively

$$I_{\operatorname{Sym}^2 H}, \quad 1_H \otimes T, \quad \operatorname{Sym}^2 T.$$

Since $\det M$ is a unit times $\det T$, substitution proves (8.1). $\square$

We now specialize this exact all-corank formula to $r = 2$. Write $\Sigma_R = \Sigma_R^{(2)}$. Then $c = 2$, and after choosing bases

$$T = \begin{pmatrix} a & b \\ c & d \end{pmatrix}, \quad \delta = \det T = ad - bc.$$

Decompose the quotient multiplication tensor according to

$$\operatorname{Sym}^2 V = \operatorname{Sym}^2 H \oplus (H \otimes W) \oplus \operatorname{Sym}^2 W$$

and write

$$A_H : \operatorname{Sym}^2 H \rightarrow S_R, \quad B_H : H \otimes W \rightarrow S_R, \quad C_H : \operatorname{Sym}^2 W \rightarrow S_R.$$

Theorem 8.2 (Exact presentation on every rank-two chart). *In the preceding coordinates one has, before taking radicals,*

$$\mathcal{I}_{\widehat{D}_R} = (\delta) I_6[A_H, B_H(1_H \otimes T), C_H \operatorname{Sym}^2 T]. \quad (8.2)$$

The formula is valid at every projection-rank-two point; no injectivity, mixed-rank, tensor-rank, or simple-contact hypothesis is required.

Proof. Proposition 3.1 gives

$$\mathcal{I}_{\widehat{D}_R} = (\det M) I_6(\gamma_R \operatorname{Sym}^2 M).$$

For $M = I_H \oplus T$, the induced map on the three summands of $\operatorname{Sym}^2(H \oplus W)$ is respectively

$$I_{\operatorname{Sym}^2 H}, \quad 1_H \otimes T, \quad \operatorname{Sym}^2 T.$$

Substitution gives (8.2). All row and column operations used to reach the block form are invertible, so the maximal-minor ideal is unchanged. $\square$

Thus the exceptional rank-two locus is included in the exact presentation. Determining radicals, associated primes, and nilpotent layers of its residual factor is a separate problem. The mixed-regular case and its first tensor-rank boundary are solved below and in Section 9.

Definition 8.3. A pair (R, H) is *mixed-regular* if A_H is injective and, after quotienting S_R by its image, the induced map

$$\overline{B}_H : H \otimes (V/H) \longrightarrow S_R / \text{im } A_H$$

is surjective with one-dimensional kernel generated by a rank-two tensor. No hypothesis is imposed on the multiplicities of the four ramification contacts.

These are precisely the first three conditions in Definition 7.1; the discriminant condition has been removed. On this locus the same Gaussian elimination as in Lemma 7.3 is valid and yields

$$\mathcal{I}_{\widehat{D}_R} = \delta J, \quad J = I_3[L(T), C T^{(2)}],$$

with $L(T)$ and $T^{(2)}$ as in (7.3).

Lemma 8.4 (Root-free residual normal form). *Over the local auxiliary coefficient ring of a mixed-regular pair,*

$$J = \delta \mathfrak{m} + (g_0, \dots, g_4), \quad \mathfrak{m} = (a, b, c, d), \quad (8.3)$$

where each g_i is homogeneous of transverse degree four. Modulo (δ) , under the Segre parametrization $T = uv^t$, their classes span

$$h_H(u) \cdot \text{Sym}^4(\mathbb{C}_v^2).$$

Here h_H is the binary quartic obtained by restricting the ramification equation to the pencil of hyperplanes containing H . The identity is valid with repeated roots, and also when the discriminant of h_H vanishes.

Proof. The four 3-minors of the linear block $L(T)$ generate $\delta \mathfrak{m}$, exactly as in Lemma 7.4. Modulo δ , write $T = uv^t$. The two pairs of linear columns are scalar multiples, by v_1, v_2 , of the two mixed columns determined by u , while the three quadratic columns are scalar multiples, by $v_1^2, v_1 v_2, v_2^2$, of $C(u^2)$. A minor containing one column from each linear pair and one quadratic column is therefore $h_H(u)$ times a degree-four monomial in v ; the five choices span $\text{Sym}^4(\mathbb{C}_v^2)$.

Every minor containing at least two quadratic columns vanishes on the rank-one cone, hence is divisible by the prime equation δ . Its transverse degree is at least five, so the quotient by δ lies in $\mathfrak{m}$. The remaining minors are the already listed linear and mixed ones. This proves (8.3). No factorization of h_H was used. When its roots are distinct, factorizing h_H recovers Lemma 7.4. $\square$

Theorem 8.5 (Depth two across contact collisions). *At every mixed-regular projection-rank-two point, the radical of $\mathcal{I}_{\widehat{D}_R}$ is (δ) and the nilradical $\widehat{\mathcal{N}}$ of the local failure ring satisfies*

$$\widehat{\mathcal{N}}^3 = 0, \quad \widehat{\mathcal{N}}^2 \neq 0, \quad \text{Ann}(\widehat{\mathcal{N}}^2) = \mathfrak{m}/(\delta J).$$

Consequently $\mathcal{Z}_2(\widehat{D}_R)$ is the rank-two stratum on the mixed-regular locus, even when the contact quartic has multiple roots. Moreover

$$\widehat{\mathcal{N}}^2|_{\Sigma_R} \simeq \mathcal{I}_{\Delta_R}^2|_{\Sigma_R}.$$

Proof. By Lemma 8.4, $J \subset \mathfrak{m}^3$ and $\delta \mathfrak{m} \subset J$. Since $\delta \in \mathfrak{m}^2$, we have $\delta^2 \in J$, hence $\delta^3 \in \delta J$. On the other hand $\delta \notin J$ by transverse degree, so $\delta^2 \notin \delta J$. Since $\delta J \subset (\delta)$ and a power of δ belongs to δJ , the radical is exactly (δ) . Thus the nilradical is generated by the class of δ and has exact index three.

For the annihilator, cancellation in the regular local coefficient ring gives

$$(\delta J : \delta^2) = (J : \delta).$$

The inclusion $\mathfrak{m} \subset (J : \delta)$ follows from $\delta \mathfrak{m} \subset J$. Conversely, use the transverse grading: J has no component of transverse degree two, whereas δ has degree two. If the transverse degree-zero part of r were nonzero, the degree-two part of $r\delta$ could not lie in J . Hence $r \in \mathfrak{m}$, proving $(J : \delta) = \mathfrak{m}$.

Finally δ is a local equation of the Schubert divisor $\hat{\Delta}_R$. Therefore the generator δ^2 of the square of the nilradical transforms as the square of the conormal line. This gives the displayed line bundle after restriction to the rank-two stratum. $\square$

We next globalize the four contacts without labelling them. Put

$$G = \text{Gr}(2, V)$$

and let $\mathcal{Q}$ be its universal rank-two quotient bundle. The quartic equation F_R of $Y_R \subset \mathbb{P}(V^*)$ restricts to every line of hyperplanes containing H . Equivalently it defines a global binary quartic

$$h_R \in H^0(G, \text{Sym}^4 \mathcal{Q}). \quad (8.4)$$

Define the contact incidence

$$\mathcal{R}_R = \{(H, [\alpha]) \in G \times Y_R : H \subset \ker \alpha\}. \quad (8.5)$$

Proposition 8.6 (The global contact cover). *The projection $\mathcal{R}_R \rightarrow Y_R$ is a $\mathbb{P}^2$ -bundle; in particular $\mathcal{R}_R$ is smooth and irreducible. Over the open set $G_R^{\text{fin}} \subset G$ on which the binary quartic (8.4) is not identically zero, the map*

$$\mathcal{R}_R \longrightarrow G$$

is finite flat of degree four. There is a globally defined discriminant divisor

$$\mathfrak{D}_R = V(\text{Disc}(h_R)), \quad \text{Disc}(h_R) \in H^0(G, (\det \mathcal{Q})^{12}).$$

Whenever this section is not identically zero, $[\mathfrak{D}_R] = 12H$ in the Pluecker class group. On the finite-flat locus G_R^{fin} , the support of the branch divisor is the restriction of $\mathfrak{D}_R$; away from $\mathfrak{D}_R$ the degree-four cover is finite etale. The monodromy action on the four sheets over that etale locus is transitive.

Proof. For $[\alpha] \in Y_R$, the fibre of $\mathcal{R}_R \rightarrow Y_R$ consists of the two-planes in the three-dimensional hyperplane $\ker \alpha$, hence is $\text{Gr}(2, \ker \alpha) \simeq \mathbb{P}^2$. These fibres form the relative Grassmannian of the rank-three hyperplane bundle on Y_R , proving smoothness and irreducibility.

For fixed H , the other projection has fibre $Y_R \cap \mathbb{P}(H^\perp)$, the zero scheme of the restricted quartic. On the locus where the restriction is nonzero this is a relative effective Cartier divisor of degree four in the $\mathbb{P}^1$ -bundle $\mathbb{P}(\mathcal{Q}^*)$, hence finite flat of degree four.

The discriminant of a binary quartic is homogeneous of degree six in its coefficients and is a relative invariant of determinant weight twelve. Applied to the section of $\text{Sym}^4 \mathcal{Q}$ in (8.4), it is therefore a section of $(\det \mathcal{Q})^{12}$. Since $\det \mathcal{Q}$ is the Pluecker line bundle up to the constant factor $\det V$, its zero divisor has class $12H$ whenever it is nonzero. On G_R^{fin} the fibre is a degree-four Cartier divisor on a projective line, so its non-etale locus is exactly the multiple-root locus, i.e. the restriction of the discriminant locus. The total incidence is irreducible; therefore the monodromy action of the etale restriction on a geometric fibre is transitive. $\square$

For R in the nonempty open $G_4^\dagger$ of Lemma 7.2, the discriminant in Proposition 8.6 is not identically zero. Let $G_R^{\text{mr}} \subset G$ denote the mixed-regular locus of Definition 8.3. The contact sheets themselves split etale locally over $G_R^{\text{fin}} \setminus \mathfrak{D}_R$. Their interpretation as the four primary ideals P_i of Theorem 7.5, however, also requires the independent mixed-regular rank conditions. Thus the exact locus for that primary decomposition is

$$G_R^{\text{mr}} \cap (G_R^{\text{fin}} \setminus \mathfrak{D}_R),$$

not the discriminant complement by itself. On this intersection the P_i are the four etale contact sheets and their permutation is the monodromy of the canonical cover (8.5).

The collision divisor has the five nonzero fibre types corresponding to the partitions

$$1 + 1 + 1 + 1, \quad 2 + 1 + 1, \quad 2 + 2, \quad 3 + 1, \quad 4$$

of the degree of a binary quartic, together with the separate containment locus $h_H \equiv 0$. Theorem 8.5 does not change across these collision types. What changes is the primary splitting of the contact part, so we do not force the simple-root intersection formula onto the boundary.

The ordinary fifth power of the rank-two Schubert ideal is of course globally defined. The stronger assertion that it remains the actual $\mathfrak{m}$ -primary component of the failure ideal through every mixed-regular collision, including the zero-quartic containment locus, is proved in Theorem 9.2; it is not inferred merely from the existence of the global ideal. The same theorem gives the exact collided primary types and separates the finite contact geometry from the containment geometry.

Corollary 8.7 (Rank-two part of Theorem 1.2). *The contact-incidence and collision assertions of Theorem 1.2 hold on the stated mixed-regular locus.*

Proof. Theorem 8.5 identifies depth two and its line bundle, while Proposition 8.6 globalizes the contact incidence and discriminant. The global residual-colon filtration, the simultaneous identification of the first two graded pieces, the collided primary components, the containment locus, and the decomposable-kernel wall are proved in Section 9. $\square$

9. THE STRATIFIED NILPOTENT ALGEBRA AT PROJECTION CORANK TWO

The root-free formula of Lemma 8.4 contains more scheme structure than the nilpotency-index statement extracted from it in the preceding revision. We record that structure before passing to the exceptional mixed-kernel boundary.

Theorem 9.1 (Global residual-colon filtration). *Let $X = X_R$, let $\widehat{\Delta}_R \subset X$ be the reduced Schubert divisor, and put*

$$\mathcal{J}_R^{\text{res}} = (\mathcal{I}_{\widehat{\Delta}_R} : \mathcal{I}_{\widehat{\Delta}_R}) \subset \mathcal{O}_X.$$

Then

$$\mathcal{I}_{\widehat{\Delta}_R} = \mathcal{I}_{\widehat{\Delta}_R} \mathcal{J}_R^{\text{res}}. \quad (9.1)$$

For every $j \geq 1$,

$$\text{Ann}_{\mathcal{O}_{\widehat{\Delta}_R}}(\widehat{\mathcal{N}}_R^j) = \frac{(\mathcal{J}_R^{\text{res}} : \mathcal{I}_{\widehat{\Delta}_R}^{j-1})}{\mathcal{I}_{\widehat{\Delta}_R}}. \quad (9.2)$$

Define the closed subscheme of the reduced divisor

$$W_j = V_{\widehat{\Delta}_R} \left((\mathcal{J}_R^{\text{res}} : \mathcal{I}_{\widehat{\Delta}_R}^{j-1}) + \mathcal{I}_{\widehat{\Delta}_R} \right). \quad (9.3)$$

If

$$\mathcal{L} = \mathcal{I}_{\widehat{\Delta}_R} / \mathcal{I}_{\widehat{\Delta}_R}^2$$

is the conormal line of the Schubert divisor, then there are canonical isomorphisms

$$\widehat{\mathcal{N}}_R^j / \widehat{\mathcal{N}}_R^{j+1} \simeq \mathcal{L}^{\otimes j} \otimes_{\mathcal{O}_{\widehat{\Delta}_R}} \mathcal{O}_{W_j}. \quad (9.4)$$

Consequently

$$\text{gr}_{\widehat{\mathcal{N}}_R} \mathcal{O}_{\widehat{\Delta}_R} \simeq \mathcal{O}_{\widehat{\Delta}_R} \oplus \bigoplus_{j \geq 1} \mathcal{L}^{\otimes j} \otimes \mathcal{O}_{W_j}, \quad (9.5)$$

with the multiplication induced by multiplication of powers of $\mathcal{I}_{\widehat{\Delta}_R}$. The sum is finite.

These formulas hold on the whole Grassmannian, including every projection corank and every rank-drop locus. On a Schur chart with local equation $d = \det T$ and residual ideal J , they become

$$\text{Ann}(N^j) = (J : d^{j-1}) / (dJ), \quad N^j / N^{j+1} \simeq A / ((d) + (J : d^{j-1})). \quad (9.6)$$

In particular the nilpotency index at the point is

$$1 + \min\{k \geq 0 : d^k \in J\}. \quad (9.7)$$

Proof. Proposition 3.1 shows locally that $\mathcal{I}_{\widehat{D}_R} = dJ$, where d is a local equation of the Cartier divisor $\widehat{\Delta}_R$. Since d is a non-zero-divisor in the regular local ring of X ,

$$(dJ : d) = J.$$

Thus J is the local expression of the intrinsic colon $\mathcal{J}_R^{\text{res}}$, and the local identities glue to (9.1).

In $A/(dJ)$ the nilradical is the image of (d) , because the reduction is $A/(d)$. Therefore

$$\text{Ann}(N^j) = (dJ : d^j)/(dJ) = (J : d^{j-1})/(dJ),$$

which proves (9.2). For the graded piece, the map

$$A \longrightarrow N^j/N^{j+1}, \quad r \longmapsto rd^j$$

has kernel

$$(d) + (J : d^{j-1}).$$

Indeed

$$rd^j \in d^{j+1} + dJ \iff rd^{j-1} \in d^j + J,$$

and the latter is equivalent to writing $r = ds + t$ with $td^{j-1} \in J$. This proves the local form of (9.4). On overlaps local equations of the Cartier divisor differ by units, and d^j transforms by the j -th tensor power of the conormal transition function. Hence the local identifications glue to (9.4) and (9.5). Nilpotence of the nilradical makes the sum finite, and the same cancellation gives (9.7). $\square$

The theorem converts the globally defined annihilator filtration into an exact residual-colon filtration. What remains is to identify the colon schemes W_j geometrically on the successive rank strata. For projection corank one, $W_1 = E_R$ and all higher W_j are empty. The next two results compute W_1, W_2 through the entire mixed-regular rank-two boundary and then across its first tensor-rank wall.

Work on a mixed-regular rank-two chart. Let A_0 be the regular auxiliary coefficient ring, put

$$P = A_0[a, b, c, d], \quad \mathfrak{m} = (a, b, c, d), \quad \delta = ad - bc,$$

and localize, or complete, at the point under consideration. Thus

$$\mathcal{I}_{\widehat{D}_R} = \delta J, \quad J = \delta \mathfrak{m} + (g_0, \dots, g_4),$$

where the g_i have transverse degree four. Define the *unsplit contact ideal*

$$I_H = (\delta, g_0, \dots, g_4). \tag{9.8}$$

Modulo δ , the Segre parametrization $T = uv^t$ identifies the homogeneous coordinate ring with

$$P/(\delta) \simeq \bigoplus_{n \geq 0} \text{Sym}^n(\mathbb{C}_u^2) \otimes \text{Sym}^n(\mathbb{C}_v^2),$$

and the image of I_H in degree n is zero for $n < 4$ and

$$h_H \text{Sym}^{n-4}(\mathbb{C}_u^2) \otimes \text{Sym}^n(\mathbb{C}_v^2) \quad (n \geq 4). \tag{9.9}$$

Theorem 9.2 (Saturated contact ideal and the fixed vertex component). *At every mixed-regular rank-two point one has the exact identities*

$$J = I_H \cap \mathfrak{m}^3 \tag{9.10}$$

and

$$\mathcal{I}_{\widehat{D}_R} = \delta I_H \cap \mathfrak{m}^5. \tag{9.11}$$

If $h_H \neq 0$, the ideal I_H is $\mathfrak{m}$ -saturated. Put $\mathcal{C}_H = V(I_H)$ for the affine contact cone. Its projectivization on the Segre quadric

$$Q = \mathbb{P}_u^1 \times \mathbb{P}_v^1 \subset \mathbb{P}^3$$

is the effective Cartier divisor

$$C_H^{\text{proj}} = V(h_H) \times \mathbb{P}_v^1 \quad (9.12)$$

of class $(4, 0)$. Consequently δI_H is also $\mathfrak{m}$ -saturated, and $\mathfrak{m}^5$ in (9.11) is an actual irredundant $\mathfrak{m}$ -primary component, not merely a globally defined ideal.

If

$$h_H = \prod_{\nu=1}^s \ell_\nu^{e_\nu}, \quad \sum_\nu e_\nu = 4,$$

over an etale coefficient extension, and P_ν denotes the height-two contact prime corresponding to ℓ_ν , then

$$\text{Ass}(P/\mathcal{I}_{\widehat{D}_R}) = \{(\delta), P_1, \dots, P_s, \mathfrak{m}\}. \quad (9.13)$$

At the generic point of P_ν , choose a second regular parameter q_ν so that $P_\nu = (\delta, q_\nu)$. The P_ν -primary part of δI_H has the exact two-parameter normal form

$$Q_{\nu, e_\nu} = (\delta^2, \delta q_\nu^{e_\nu}, q_\nu^{e_\nu+1}). \quad (9.14)$$

For $e_\nu = 1$ this is P_ν^2 ; for a collided contact it is the corresponding non-ordinary primary thickening.

If $h_H \equiv 0$, then

$$I_H = (\delta), \quad J = \delta \mathfrak{m}, \quad \mathcal{I}_{\widehat{D}_R} = \delta^2 \mathfrak{m} = (\delta^2) \cap \mathfrak{m}^5. \quad (9.15)$$

In particular the rank-two vertex component remains $\mathfrak{m}^5$ on the containment locus.

Proof. The equality (9.10) is a graded calculation. The ideal I_H is generated in transverse degrees two and four, its only degree-two generator being δ . Intersecting with $\mathfrak{m}^3$ therefore removes exactly the scalar multiple of δ and leaves $\delta \mathfrak{m} + (g_0, \dots, g_4) = J$.

Since the initial form of δ has degree two and is nonzero in the polynomial associated graded ring,

$$(\mathfrak{m}^5 : \delta) = \mathfrak{m}^3.$$

Cancellation in the regular ring gives

$$\delta I_H \cap \mathfrak{m}^5 = \delta(I_H \cap (\mathfrak{m}^5 : \delta)) = \delta(I_H \cap \mathfrak{m}^3) = \delta J,$$

which proves (9.11).

Assume first $h_H \neq 0$. Formula (9.9) is precisely the section ideal of $\mathcal{O}_Q(-4, 0) \subset \mathcal{O}_Q$, cut out by the nonzero section h_H of $\mathcal{O}_Q(4, 0)$. Hence it is saturated and defines the Cartier divisor (9.12). The inverse image $I_H \subset P$ is therefore $\mathfrak{m}$ -saturated. The product δI_H is saturated as well: if $\mathfrak{m}^N f \subset \delta I_H$, choose a coordinate not divisible by the prime element δ to see that $\delta \mid f$; after cancelling δ , saturation of I_H applies. Thus $\mathfrak{m} \notin \text{Ass}(P/\delta I_H)$. On the other hand $\delta^2 \in \delta I_H$ has transverse degree four and does not belong to $\mathfrak{m}^5$, so the $\mathfrak{m}^5$ factor in (9.11) is irredundant.

The divisor C_H^{proj} has components $e_\nu(\{\ell_\nu = 0\} \times \mathbb{P}^1)$. Hence P/I_H is Cohen–Macaulay of pure codimension two and its associated primes are exactly the P_ν . Multiplication by δ gives an exact sequence

$$0 \longrightarrow P/I_H \xrightarrow{\delta} P/\delta I_H \longrightarrow P/(\delta) \longrightarrow 0.$$

It follows, and localization shows equality, that the associated primes of $P/\delta I_H$ are $(\delta), P_1, \dots, P_s$. Intersecting with the irredundant $\mathfrak{m}$ -primary ideal $\mathfrak{m}^5$ gives (9.13).

At the generic point of P_ν , the Cartier divisor has equation $q_\nu^{e_\nu}$, so

$$(I_H)_{P_\nu} = (\delta, q_\nu^{e_\nu}), \quad (\delta I_H)_{P_\nu} = (\delta^2, \delta q_\nu^{e_\nu}).$$

The ideal

$$(\delta^2, \delta q_\nu^{e_\nu}, q_\nu^{e_\nu+1})$$

is P_ν -primary, and its colon by δ is $(\delta, q_\nu^{e_\nu})$. Therefore

$$(\delta) \cap Q_{\nu, e_\nu} = \delta(\delta, q_\nu^{e_\nu}),$$

which proves the asserted generic primary normal form.

If $h_H = 0$, every g_i vanishes modulo δ . Since it has transverse degree four, $g_i = \delta q_i$ with $q_i \in \mathfrak{m}^2$. Thus $I_H = (\delta)$ and $J = \delta \mathfrak{m}$. Finally

$$(\delta^2) \cap \mathfrak{m}^5 = \delta^2(\mathfrak{m}^5 : \delta^2) = \delta^2 \mathfrak{m},$$

again by the polynomial associated graded ring. This is (9.15). $\square$

The preceding theorem gives the requested specialization law without labelling roots. A collision changes the Cartier multiplicities in C_H and hence the primary type (9.14); it does not change the vertex component. The zero quartic is the unique point at which the Cartier contact divisor ceases to be a divisor and fills the whole Segre quadric.

Theorem 9.3 (Associated graded nilpotent algebra across coranks one and two). *On a mixed-regular rank-two chart let*

$$A = P/(\delta J), \quad N = \sqrt{0_A}.$$

Then $N = (\delta)/(\delta J)$, $N^3 = 0 \neq N^2$, and

$$N/N^2 \simeq P/I_H, \tag{9.16}$$

$$N^2 \simeq P/\mathfrak{m}. \tag{9.17}$$

Equivalently, with $\mathcal{L}$ the conormal line of the Schubert divisor, the associated graded algebra has the intrinsic form

$$\mathrm{gr}_N \mathcal{O}_{\widehat{D}_R} \simeq \mathcal{O}_{\widehat{\Delta}_R} \oplus (\mathcal{L} \otimes \mathcal{O}_{C_H}) \oplus (\mathcal{L}^{\otimes 2} \otimes \mathcal{O}_{\Sigma_R}) \tag{9.18}$$

on the mixed-regular neighbourhood, with multiplication induced by the incidence of the affine contact cone $\mathcal{C}_H$ with the rank-two vertex Σ_R . Moreover

$$\mathrm{Ann}_A(N) = J/(\delta J), \quad \mathrm{Ann}_A(N^2) = \mathfrak{m}/(\delta J), \tag{9.19}$$

so

$$\mathcal{Z}_1 = V(I_H \cap \mathfrak{m}^3), \quad \mathcal{Z}_2 = V(\mathfrak{m}).$$

On the punctured determinant cone, i.e. on the adjacent projection-corank-one locus, $\mathcal{Z}_1$ is exactly the ramification support E_R . Thus $\mathcal{C}_H$ is its scheme-theoretic closure away from the vertex, while $\mathcal{Z}_1$ records in addition the canonical third-order vertex thickening forced by the Fitting ideal.

Proof. The radical of δJ is (δ) , so the nilradical is generated by the class of δ . The isomorphism (9.16) follows by cancelling one factor:

$$N/N^2 = \frac{(\delta)}{(\delta^2) + (\delta J)} \simeq \frac{P}{(\delta) + J} = \frac{P}{I_H}.$$

For the square, multiplication by δ^2 identifies its kernel with

$$(\delta J : \delta^2) = (J : \delta) = \mathfrak{m}$$

by Theorem 8.5; hence (9.17). The transition functions of the local generator δ are those of the conormal line, giving (9.18).

Cancellation also gives

$$(\delta J : \delta) = J,$$

while the preceding colon calculation gives the second formula in (9.19). Away from the vertex $\mathfrak{m}$ becomes the unit ideal, so $J = I_H$. On the smooth rank-one part of the determinant divisor, the corank-one normal form $t(t, f)$ identifies this contact equation with the ramification equation. Hence the restriction is precisely E_R . $\square$

We finally cross the first non-mixed-regular wall. This is the rank-one tensor wall in the one-dimensional mixed kernel, explicitly requested by the referee.

Definition 9.4. Assume A_H is injective and $\overline{B}_H$ is surjective with one-dimensional kernel generated by a decomposable tensor $h_1 \otimes w_1$. Choose h_2, w_2 completing bases and put

$$q_1 = \overline{B}_H(h_1 \otimes w_2), \quad q_2 = \overline{B}_H(h_2 \otimes w_1).$$

We call the point *transverse decomposable-kernel regular* if

$$\det[q_1, q_2, \overline{C}_H(w_1^2)] \neq 0.$$

This condition is independent of the chosen completing vectors. It says that the contact root forced by the decomposable kernel is simple in the direction transverse to the kernel wall.

Theorem 9.5 (The decomposable mixed-kernel wall). *At a transverse decomposable-kernel regular rank-two point there are bases for which*

$$L_{\text{dec}}(T) = \begin{pmatrix} c & d & 0 & 0 \\ 0 & 0 & a & b \\ 0 & 0 & c & d \end{pmatrix}. \quad (9.20)$$

Let

$$J_{\text{dec}} = I_3[L_{\text{dec}}(T), C T^{(2)}], \quad K_{\text{dec}} = (c, d, a^2, ab, b^2).$$

Then

$$(J_{\text{dec}} : \delta) = K_{\text{dec}}. \quad (9.21)$$

Hence the local failure ring still has nilradical of exact index three, but its second intrinsic depth support is no longer reduced:

$$\mathcal{Z}_2 = V(K_{\text{dec}}). \quad (9.22)$$

Its support is the rank-two vertex, and its transverse local length is three. Thus crossing the tensor-rank-two wall changes the scheme structure of $\mathcal{Z}_2$ before it changes its set-theoretic support.

Proof. The stated normal form is obtained by taking $\ker \overline{B}_H = \mathbb{C}(h_1 \otimes w_1)$ and using

$$q_1, q_2, q_3 = \overline{B}_H(h_2 \otimes w_2)$$

as target basis. Applying $1_H \otimes T$ gives (9.20). Its maximal minors generate

$$I_3(L_{\text{dec}}) = \delta(c, d).$$

After replacing the first quadratic target column by its component modulo $\langle q_1, q_2 \rangle$, the transversality assumption allows us to normalize $\overline{C}_H(w_1^2) \equiv q_3$. The minors containing one column from each mixed group and this quadratic column give

$$\delta a^2, \quad \delta ab, \quad \delta b^2 \in J_{\text{dec}}.$$

Together with $I_3(L_{\text{dec}})$ this proves $\delta K_{\text{dec}} \subset J_{\text{dec}}$, hence $K_{\text{dec}} \subset (J_{\text{dec}} : \delta)$.

For the reverse inclusion use the transverse grading. The degree-three part of J_{dec} is exactly $\delta(c, d)$; every minor using a quadratic column has degree at least four. Modulo K_{dec} , every class is represented by

$$\lambda + \mu a + \nu b.$$

If $r\delta \in J_{\text{dec}}$, the degree-two term rules out $\lambda \neq 0$, while the degree-three identity

$$(\mu a + \nu b)\delta \in \delta(c, d)$$

forces $\mu = \nu = 0$. Thus $r \in K_{\text{dec}}$, proving (9.21).

Since $\delta \in (c, d)$, one has $\delta^2 \in \delta(c, d) \subset J_{\text{dec}}$, whereas $\delta \notin J_{\text{dec}}$ by degree. Therefore the nilradical of $P/(\delta J_{\text{dec}})$ has exact index three. Finally

$$\text{Ann}(N^2) = (\delta J_{\text{dec}} : \delta^2) = (J_{\text{dec}} : \delta) = K_{\text{dec}}.$$

The quotient by K_{dec} has basis $1, a, b$, proving the length assertion. $\square$

Proposition 9.6 (The containment locus is the finite line scheme). *For fixed $R \in G_4^\circ$, the locus*

$$\mathcal{C}_R = \{H \in \text{Gr}(2, V) : h_H \equiv 0\}$$

is canonically the Fano scheme $F_1(Y_R)$ of lines on the smooth quartic Y_R . It is finite and reduced. At its mixed-regular points the local failure ideal is (9.15), and the contact morphism $\mathcal{R}_R \rightarrow \text{Gr}(2, V)$ has fibre $\mathbb{P}^1$, rather than a finite length-four fibre.

Proof. The line of hyperplanes containing H is $\mathbb{P}(H^\perp) \subset \mathbb{P}(V^*)$. Its quartic restriction is h_H , so $h_H = 0$ identically exactly when $\mathbb{P}(H^\perp) \subset Y_R$. This identifies $\mathcal{C}_R$ with $F_1(Y_R)$.

If $L \subset Y_R$ is such a line, the normal sequence in $\mathbb{P}^3$ gives

$$\deg N_{L/Y_R} = \deg N_{L/\mathbb{P}^3} - \deg N_{Y_R/\mathbb{P}^3}|_L = 2 - 4 = -2.$$

Since N_{L/Y_R} has rank one, $N_{L/Y_R} \simeq \mathcal{O}_{\mathbb{P}^1}(-2)$, and therefore $H^0(N_{L/Y_R}) = 0$. The Zariski tangent space of the Fano scheme vanishes at every point. The Fano scheme is projective and zero-dimensional; a zero-dimensional local algebra with zero tangent space is a field by Nakayama. Hence it is finite reduced. The final two assertions follow from Theorem 9.2 and the definition of the incidence fibre. $\square$

10. THE CANONICAL REES DEGENERATION AND NORMAL CONE

We pass from the associated graded algebra to the specialization mechanism that produces it. This places every projection rank in one special fibre over the whole Schubert divisor, while the construction itself is attached to the abstract nonreduced scheme.

Proof of Theorem 1.5. The nilradical $\widehat{\mathcal{N}}_R$ is intrinsic, coherent, and nilpotent. On an affine open $\text{Spec } A \subset \widehat{D}_R$, write $N \subset A$ for its nilradical and set

$$\mathcal{R}_N = A[t, Nt^{-1}] \subset A[t, t^{-1}].$$

This is the subalgebra generated by $A[t]$ and the elements nt^{-1} , $n \in N$. Formation of the nilradical and its powers commutes with localization, so these affine algebras glue.

Multiplication by t is injective on $\mathcal{R}_N$, since it is a subring of $A[t, t^{-1}]$. Thus every stalk is torsion-free as a $\mathbb{C}[t]$ -module. Since $\mathbb{C}[t]$ is a principal ideal domain, torsion-free $\mathbb{C}[t]$ -modules are flat. Hence the resulting scheme is flat over $\mathbb{A}^1$.

After inverting t ,

$$\mathcal{R}_N[t^{-1}] = A[t, t^{-1}],$$

so the restriction over $\mathbb{G}_m$ is $\widehat{D}_R \times \mathbb{G}_m$. For the special fibre, the extended-Rees calculation gives

$$\mathcal{R}_N/(t) \simeq \bigoplus_{j \geq 0} N^j / N^{j+1} = \text{gr}_N A.$$

Indeed the degree- j class is represented by nt^{-j} , $n \in N^j$, and vanishes modulo t precisely when $n \in N^{j+1}$. These identifications commute with localization. Consequently

$$\mathfrak{D}_{R,0} \simeq \text{Spec}_{\widehat{\Delta}_R} \text{gr}_{\widehat{\mathcal{N}}_R} \mathcal{O}_{\widehat{D}_R}.$$

Because $\widehat{\mathcal{N}}_R$ is the ideal of $\widehat{\Delta}_R = (\widehat{D}_R)_{\text{red}}$ in $\widehat{D}_R$, the right-hand side is the normal cone $C_{\widehat{\Delta}_R/\widehat{D}_R}$.

Theorem 9.1 identifies each graded piece with

$$\mathcal{L}^{\otimes j} \otimes \mathcal{O}_{W_j},$$

and its multiplication with multiplication of successive powers of the Schubert ideal, equivalently with the compatible maps in the colon tower. An abstract scheme isomorphism preserves the

nilradical, so it preserves the extended Rees algebra, the family, and its special fibre. This proves functoriality. $\square$

Corollary 10.1 (One specialization across the first boundary). *The canonical special fibre restricts as follows.*

- (1) *On the adjacent projection-corank-one locus it truncates after degree one, whose support is E_R .*
- (2) *On the mixed-regular rank-two locus it is*

$$\mathcal{O}_{\widehat{\Delta}_R} \oplus (\mathcal{L} \otimes \mathcal{O}_{\mathcal{C}_H}) \oplus (\mathcal{L}^{\otimes 2} \otimes \mathcal{O}_{\Sigma_R}).$$

Contact collisions alter the Cartier multiplicities in $\mathcal{C}_H$ but leave the vertex-primary factor present.

- (3) *At a transverse decomposable mixed-kernel wall, the degree-two support is*

$$V(c, d, a^2, ab, b^2),$$

a length-three thickening of the rank-two vertex.

These are restrictions of one canonical normal cone.

Proof. The first assertion is the corank-one normal form $t(t, f)$. The second follows from Theorems 9.2 and 9.3; the third is Theorem 9.5. The preceding theorem places all three in the same special fibre. $\square$

Remark 10.2 (Presentation versus structure). The residual block identity of Theorem 8.1 is a presentation on a chosen Schur chart. By contrast, $\mathfrak{D}_R \rightarrow \mathbb{A}^1$ is built from the intrinsic nilradical. Its special fibre records every annihilator layer and every multiplication map between successive layers. Thus the Rees family supplies the global specialization theorem requested by the higher-corank objection; the block formula remains a tool for computing particular strata of that object.

11. THE CLASSICAL INCIDENCE SURFACE AND THE NEW INVARIANT

This section distinguishes three geometric objects associated with the same relation system. Let $e = 4$ and let $q_i(x, y)$ denote the symmetric bilinear polarization of $q_i(x)$, normalized by $q_i(x, x) = q_i(x)$. Define

$$Z_R = \{([x], [y]) \in \mathbb{P}(V^*) \times \mathbb{P}(V^*) : q_i(x, y) = 0 \text{ for all } i\}. \quad (11.1)$$

Proposition 11.1. *For $R \in G_4^\circ$, projection on the first factor identifies Z_R with Y_R . Transposition of the two factors induces a fixed-point-free involution on this K3 surface. The polarization recovered in Theorem 1.1 is the first projection's quartic polarization.*

Proof. The bilinear equations at x are $J_R(x)y = 0$, up to the common nonzero factor two. Off Y_R there is no projective solution; on Y_R the matrix has rank exactly three because Y_R is smooth. Inverting a nonzero three-minor gives a unique kernel line, with its scheme structure, and these kernel lines glue to an inverse for the projection $Z_R \rightarrow Y_R$. The equations are symmetric in x, y , so transposition acts. A fixed point would have $[x] = [y]$ and hence $q_i(x) = 0$ for every i , contrary to basepoint-freeness. The final assertion follows from the first projection and its hyperplane bundle. $\square$

The discriminant of the web is a different embedded hypersurface. If $q_i(x) = x^t A_i x$, its equation is

$$\det\left(\sum_{i=0}^3 \lambda_i A_i\right) = 0 \quad \text{in } \mathbb{P}(R), \quad (11.2)$$

whereas the ramification equation is $\det J_R(x) = 0$ in $\mathbb{P}(V^*)$. The former varies the quadric in the web; the latter varies a point of the source of the quadratic morphism. The two determinants must not be identified simply because both have degree four. The classical construction recalled in [13, §4] relates the bilinear K3 surface to the resolution of the web's symmetroid and to a nodal Enriques quotient; see also the account of Reye congruences in [9]. Proposition 11.1 places our ramification surface in that construction by an explicit kernel-line isomorphism. No claim is made that the existence of that K3 surface, its involution, or webs of quadrics is new.

The invariant used here starts instead with the abstract multiplication-failure scheme. It obtains E as the annihilator of its nilradical, constructs $\omega_E \otimes \mathcal{N}^{-9}$ without the web or either projective space, and recovers the section ring of $\mathcal{O}_{Y_R}(4)$. The corank-two theorem further recovers an additional associated support from the square of the nilradical. These are statements about nonreduced multiplication-failure schemes, not merely about the classical incidence surface or the discriminant cycle.

12. REYE GEOMETRY, INFINITESIMAL MODULI, AND THE NOVELTY BOUNDARY

We now separate three statements that were too close together in the preceding revision: the classical moduli of webs and Reye congruences, the exact differential calculation for the polarized K3 map, and the nonreduced reconstruction theorem.

12.1. The classical nine-dimensional quotient. For $e = 4$, a relation space $R \subset \text{Sym}^2 V$ is a web of quadrics. The bilinear incidence surface Z_R of §11 is a K3 surface with its fixed-point-free factor-exchange involution; its quotient is the classical Reye Enriques surface associated with the web. This construction and the resulting congruence of lines are classical; see [7, 1, 10, 13, 9, 16].

Arrondo–Sols compute that the Hilbert scheme of Reye congruences is smooth of dimension 24 and rational, and describe it by an open Grassmannian parameter of the corresponding data. After quotienting by the 15-dimensional projective coordinate group, the expected moduli dimension is therefore nine. In characteristic zero the nodal Enriques locus is the classical nine-dimensional divisor in the ten-dimensional Enriques moduli space; the modern theorem [16] identifies every classical nodal Enriques surface as a Reye congruence.

Thus the equality

$$24 - 15 = 9$$

is not a novelty claim of this paper.

Proposition 12.1 (The classical tangent quotient). *At a general regular web with finite projective stabilizer, the tangent space to the Reye moduli locus is the quotient of the 24-dimensional web tangent space by the 15-dimensional infinitesimal $\text{PGL}(V)$ -orbit:*

$$T_{[R]}\mathcal{M}_{\text{Reye}} \simeq T_R \text{Gr}(4, \text{Sym}^2 V) / \mathfrak{pgl}(V), \quad \dim T_{[R]}\mathcal{M}_{\text{Reye}} = 9.$$

The differential of the polarized-K3 map of Theorem 1.4 factors through this quotient.

Proof. The first statement is the tangent-space form of the smooth 24-dimensional Hilbert calculation of Arrondo–Sols on the regular web locus, followed by the infinitesimal quotient by projective coordinates. The stabilizer is finite at a general point, so the orbit tangent has dimension 15.

The construction $R \mapsto Z_R \simeq Y_R$ is equivariant for $\text{PGL}(V)$. Passing from an embedded quartic to its polarized isomorphism class kills the infinitesimal projective orbit; hence the differential factors through the displayed quotient. $\square$

Proposition 12.2 (Web parameter and Reye Hilbert point). *Let*

$$\mathcal{G} = \text{Gr}(4, \text{Sym}^2 V)$$

and let $\mathcal{G}^{\text{reg}}$ be the open on which the classical Reye construction has the expected Hilbert polynomial. The assignment of a web to its Reye congruence is a $\text{PGL}(V)$ -equivariant morphism

$$\Phi : \mathcal{G}^{\text{reg}} \longrightarrow \text{Hilb}_{\text{Reye}}.$$

Under the Grassmannian parameterization of the regular Reye locus in Arrondo–Sols, Φ identifies a dense open of $\mathcal{G}^{\text{reg}}$ with the corresponding 24-dimensional Hilbert component. In particular, at a general regular web,

$$d\Phi_R : \text{Hom}(R, \text{Sym}^2 V/R) \xrightarrow{\sim} T_{[\text{Reye}(R)]} \text{Hilb}_{\text{Reye}}.$$

After quotienting by the infinitesimal projective orbit this gives the tangent quotient in Proposition 12.1.

Proof. Over $\mathcal{G}$ the universal rank-four subbundle of $\text{Sym}^2 V$ gives the universal web. The incidence equations defining the Reye congruence are polynomial in the universal Pluecker coordinates. Restrict to the open on which their Hilbert polynomial is the regular Reye polynomial. Flatness there and the Hilbert functor give the morphism Φ ; equivariance is immediate from the construction.

Arrondo–Sols describe the regular Reye Hilbert component by the open Grassmannian parameter of the defining four-dimensional system of quadrics. With the present convention this parameter is precisely a dense open of $\text{Gr}(4, \text{Sym}^2 V)$ (duality only changes whether one writes the system or its annihilator). Their smoothness and dimension-24 statement therefore identifies the tangent space of the Hilbert point with the tangent space of this Grassmannian. The latter is $\text{Hom}(R, \text{Sym}^2 V/R)$, proving the displayed isomorphism. Since the construction is $\text{PGL}(V)$ -equivariant, the orbit tangent maps to the orbit tangent and the quotient gives Proposition 12.1. $\square$

The role of Lemma 6.1 is now precise. It is not used to discover the number nine. Its exact 25×25 integral minor proves that the particular polarized K3 attached to our multiplication problem does not collapse any of the 24 embedded web directions at the explicit rational point. After projective quotient, it proves rank nine for the map to polarized K3 moduli. Since maximal rank is open, the induced map from the classical nine-dimensional Reye deformation space to our polarized K3 image is generically unramified. This is the conceptual tangent statement behind Theorem 1.4; the determinant is an exact arithmetic witness for it.

12.2. What the finite ambiguity means. The preceding paragraph also locates the remaining Torelli ambiguity. The failure scheme first reconstructs the polarized K3 $(Y_R, \mathcal{O}_{Y_R}(1))$. A relation system additionally supplies the free involution coming from factor exchange on the bilinear incidence model and hence an Enriques quotient with its Reye data. Classical Reye geometry then has substantial rigidity. In particular, Dolgachev–Reider prove that for a Reye polarization on an Enriques surface the stable rank-two bundle with c_1 equal to that polarization and $c_2 = 3$ is the Reye bundle and is unique up to isomorphism; it is the restriction of the universal quotient bundle from $G(2, 4)$.

Consequently the generic finite fibre in Theorem 1.4 should not be read as an uncontrolled family of quadratic tensors. Once a compatible Enriques quotient and Reye polarization are fixed, the Grassmannian Reye bundle is rigid in the preceding sense. The remaining finite packet is concentrated in the finite choices of compatible Reye/Enriques data over the recovered polarized K3, together with the passage back to the web. The present paper proves finiteness but does not claim a numerical degree or generic degree one; such a claim would require a separate enumeration of those compatible involutions and web structures.

There is a second useful localization. On the mixed-regular projection-corank-two locus, the unsplit contact quartic of §8 is the restriction of the already recovered quartic Y_R to the pencil of hyperplanes containing H . Thus the generic corank-two contact cover itself is determined by

the polarized K3. The following theorem turns the finite Torelli packet into a precise finite-cover statement and identifies its monodromy.

Theorem 12.3 (Generic Torelli packet as a finite etale monodromy cover). *Let $\mathcal{W}$ be the irreducible nine-dimensional coarse quotient of the regular web locus by projective equivalence on the finite-stabilizer open, and let $\mathcal{K}$ be the nine-dimensional polarized-K3 image of Theorem 1.4. There are nonempty opens*

$$\mathcal{W}^\circ \subset \mathcal{W}, \quad \mathcal{K}^\circ \subset \mathcal{K}$$

such that

$$\tau : \mathcal{W}^\circ \longrightarrow \mathcal{K}^\circ$$

is finite etale of a constant degree

$$d_{\text{Tor}} = [\mathbb{C}(\mathcal{W}) : \mathbb{C}(\mathcal{K})].$$

Its geometric monodromy is transitive on the d_{Tor} points of a fibre. Equivalently, if L is the normal closure of $\mathbb{C}(\mathcal{W})/\mathbb{C}(\mathcal{K})$, the monodromy group is the permutation image of

$$\text{Gal}(L/\mathbb{C}(\mathcal{K}))$$

on the embeddings of $\mathbb{C}(\mathcal{W})$ in L .

The degree-four contact incidence and its discriminant are pulled back from the polarized quartic on $\mathcal{K}^\circ$. Hence they are constant on a Torelli fibre and do not account for d_{Tor} ; any further separation of the finite packet must use web-dependent structure beyond the polarized quartic, such as the exceptional mixed-kernel or deeper projection-corank strata.

Proof. The source $\mathcal{W}$ is irreducible because it is obtained from a nonempty open of the irreducible Grassmannian $\text{Gr}(4, \text{Sym}^2 V)$ by the projective-coordinate quotient on the finite-stabilizer locus. Theorem 1.4 says that the dominant map $\tau : \mathcal{W} \dashrightarrow \mathcal{K}$ is generically finite. Thus the induced extension of function fields is finite. In characteristic zero it is separable.

Normalize a nonempty affine open of $\mathcal{K}$ in $\mathbb{C}(\mathcal{W})$. After deleting the complement of the locus on which the original rational map is defined, and then deleting the branch and nonflat loci, one obtains a finite etale morphism from a nonempty open of $\mathcal{W}$. Its degree is the function-field degree displayed above. Connectedness of the generic source, or equivalently irreducibility of $\mathcal{W}$, gives transitivity of geometric monodromy. The description by the normal closure is the standard field-theoretic realization of that action.

The contact incidence is defined solely from the recovered embedded quartic $Y \subset \mathbb{P}^3$ by

$$(H, [\alpha]) \quad \text{with} \quad H \subset \ker \alpha,$$

and the discriminant is the discriminant of its restricted binary quartic. Both constructions commute with isomorphism of the polarized quartic. They therefore descend to $\mathcal{K}^\circ$ and pull back along τ , proving the last assertion. $\square$

12.3. Comparison with nonreduced reconstruction theory. The use of nilpotent layers has classical antecedents. A ribbon is a square-zero double structure with an invertible conormal module, and Bayer–Eisenbud developed its intrinsic and canonical geometry [4]. Manolache surveys Cohen–Macaulay nilpotent structures and their filtrations [15]. Primitive multiple schemes carry a canonical filtration by lower multiplicities in the sense of Drézet [11].

Our local corank-one model

$$t(t, f) = (t) \cap (t, f)^2$$

indeed has a square-zero nilradical line on the embedded support, and the corank-two model has one further nonzero nilpotent power. We do not claim that the existence of a nilpotent filtration

is new. The specific contribution is the following chain of implications, which does not occur in the cited multiple-structure theory:

$$\text{multiplication Fitting scheme} \implies (\mathcal{Z}_1, \widehat{\mathcal{N}}|_{\mathcal{Z}_1}) \implies \omega_{\mathcal{Z}_1} \otimes \widehat{\mathcal{N}}^{-9} \implies (Y_R, \mathcal{O}_{Y_R}(1)),$$

together with

$$\mathcal{Z}_2 = \text{rank-two Schubert support}$$

on the mixed-regular higher-corank locus. The ambient failure scheme is not assumed to be a primitive multiple scheme, and the filtration is defined by annihilators of powers of its intrinsic nilradical rather than by choosing an ambient thickening.

12.4. Ballico's failure-locus framework. The complete theorem text of Ballico's 1993 article [2] was not available to this revision. We therefore do not infer either anticipation or non-anticipation from its title or metadata. A theorem-level comparison is possible with Ballico's accessible 1996 continuation [3]. There the basic data are an integral projective variety X , a very ample line bundle L , the complete embedding by $H^0(X, L)$, and multiplication maps such as

$$H^0(L) \otimes H^0(L^k) \longrightarrow H^0(L^{k+1}).$$

The principal results propagate failure of quadratic or higher normality to suitable linear sections and finite subschemes; the parameter Grassmannian is a Grassmannian of projective linear sections.

The present construction has a different input and a different scheme-theoretic output. The fixed object is the finite local cube-zero algebra B_R ; a point of X_R is a unital subspace $\mathbb{C}1 \oplus K \subset B_R$; and the failure scheme is the zeroth Fitting scheme of

$$\text{Sym}^m(\mathcal{O} \oplus \mathcal{K}) \longrightarrow B_R \otimes \mathcal{O}.$$

Its embedded associated supports, nilradical powers, and the polarized K3 reconstructed from their line bundles are part of the statement. Ballico's 1996 theorems are therefore an important precedent for organizing multiplication failure over a Grassmannian, but they do not by themselves supply the Fitting normal forms or the nilpotent-depth reconstruction proved here. This comparison is deliberately limited to the theorem text actually inspected; the separate 1993 comparison remains documentary work rather than a premise in a novelty claim.

12.5. Residue of the main theorem. After these classical comparisons, the claims that remain specific to this paper are concise.

First, an abstract multiplication-failure scheme reconstructs the quartic polarization by the intrinsic line $\omega_{E_R} \otimes \mathcal{N}_R^{-9}$ and its section ring. Second, the same nonreduced scheme has a functorial annihilator filtration whose next nonzero stage recovers the first higher-corank Schubert support. Third, on the mixed-regular rank-two locus the unsplit contact ideal has a complete collision specialization law, the fifth-power vertex component remains genuinely primary, and the associated graded nilpotent algebra identifies both the contact cone and the rank-two Schubert support. The first decomposable mixed-kernel wall is also computed scheme-theoretically. Finally, Theorem 12.3 identifies the generic inverse problem with a finite étale monodromy cover. These statements are independent of whether the nine-dimensional Reye family itself is classical.

APPENDIX A. WEIGHTED DETERMINANTAL FLAGS AND THE CLASSICAL ORDER PROFILE

We record explicitly how the retained weighted-matrix calculation fits the classical determinantal theory. This comparison concerns the generic matrix only; the fixed-tensor theorem of §7 was proved independently.

Let $A = (a_{ij})$ be a generic $q \times p$ matrix over a characteristic-zero field k , with $p \geq q$, and put $R_0 = k[a_{ij}]$, $I_0(A) = R_0$. For a nonincreasing sequence $\rho = (r_1, \dots, r_s)$ write

$$I^\rho = \prod_{\nu=1}^s I_{r_\nu}(A), \quad \gamma_j(\rho) = \sum_{\nu=1}^s \max(r_\nu - j + 1, 0).$$

The classical formula is

$$I^\rho = \bigcap_{j=1}^{r_1} I_j(A)^{(\gamma_j(\rho))}. \quad (\text{A.1})$$

Here the superscript in parentheses denotes symbolic power. See De Concini–Eisenbud–Procesi [8], and its statement as Theorem 2.4 in Bruns–Conca [5], together with Proposition 2.2 there for the symbolic-order basis. The characteristic hypotheses and attribution are stated on printed pages 13–14 of that exposition.

Proposition A.1 (Coefficientwise suspension). *In $R_0[t]$, put*

$$F = t \sum_{i=0}^q t^{q-i} I_i(A), \quad P_j = (t, I_j(A)).$$

Then

$$F = (t) \cap \bigcap_{j=1}^q P_j^{(q-j+2)}. \quad (\text{A.2})$$

This is an irredundant primary decomposition. At a point with $t = 0$ where A has rank a , the nilradical of the corresponding local quotient has index $q - a + 1$.

Proof. In the regular local ring $(R_0)_{I_j}$, the symbolic order of an i -minor is $\max(i - j + 1, 0)$. Adjoining t gives the coefficient rule

$$\sum_{u \geq 0} t^u f_u \in P_j^{(b)} \iff f_u \in I_j^{(\max(b-u, 0))} \text{ for every } u. \quad (\text{A.3})$$

For precision, the associated graded ring of the maximal ideal $(t, I_j(R_0))_{I_j}$ is a polynomial ring over its residue field; its homogeneous t -degrees are independent. The order of the displayed sum is therefore the minimum of $u + \text{ord}_{I_j}(f_u)$. Contracting the maximal-ideal power proves (A.3), including the assertion that each contracted symbolic power is primary.

Intersecting with (t) deletes coefficient degree zero. For $1 \leq u \leq q$, put $k = q + 1 - u$. The remaining coefficient conditions on the right of (A.2) are

$$f_u \in \bigcap_{j=1}^k I_j^{(k-j+1)} = I_k,$$

the one-factor case $\rho = (k)$ of (A.1). For $u \geq q + 1$ there is no condition. These are exactly the coefficients of F , proving the equality.

For irredundancy, let d_{j-1} be a fixed $(j-1)$ -minor, with $d_0 = 1$. The element $t^{q-j+1} d_{j-1}^2$ belongs to every component other than $P_j^{(q-j+2)}$. Indeed, its order along P_j is $q - j + 1$, while at P_k for $k < j$ its order is $q - j + 1 + 2(j - k) \geq q - k + 2$, and for $k > j$ its t -order suffices. The element d_q^2 witnesses the necessity of (t) . The associated primes are therefore exactly the displayed radicals.

After inverting an a -minor and Schur reducing, the localized coefficient ideal contains t^{q-a+1} , whereas t^{q-a} has coefficient one in degree $q - a$, where membership would require a one-minor of the residual matrix. That coefficient ideal is proper at the point. Thus the radical (t) has the asserted exact index. $\square$

The exponent $q - j + 2$ is therefore the scalar-variable suspension of the classical minor order $k - j + 1$, taken coefficientwise. This is a direct consequence of the classical profile plus (A.3); it is not a new straightening theorem. The presentation $I_q[A, tI_q] = \sum_i t^{q-i} I_i(A)$ can also be viewed as substituting a scalar identity block in a larger generic matrix. Such substitution alone does not justify preservation of primary components or flatness. The coefficient proof is what supplies the exact intersection here, while the ideal $\delta\mathbf{m} + (g_0, \dots, g_4)$ in §7 requires its separate fixed-tensor analysis.

APPENDIX B. AN EXPLICIT SYSTEM AND EXACT CERTIFICATE SPECIFICATIONS

The rational system used throughout is the following. It is the same relation system as in the preceding manuscript's exact smoothness example, now used for a differential and a higher-corank calculation.

$$\begin{aligned} q_0 &= x_0^2 + 2x_0x_1 - 2x_0x_2 + 2x_0x_3 + x_1x_2 + x_1x_3 + x_2x_3 \\ q_1 &= -x_0x_1 - x_0x_3 + x_1^2 - x_1x_2 - x_1x_3 - 2x_2x_3 \\ q_2 &= 2x_0x_1 - x_0x_2 - 2x_0x_3 + 2x_1x_2 + x_2^2 + x_2x_3 \\ q_3 &= x_0x_2 + x_1x_2 - 2x_2x_3 + x_3^2 \end{aligned}$$

For a compact complete specification of its quartic, list the 35 degree-four exponent tuples in descending lexicographic order, starting with $(4, 0, 0, 0)$ and ending with $(0, 0, 0, 4)$. The coefficients of $F = \det(\partial q_i / \partial x_j)$ in that order are the rows of

$$\begin{pmatrix} -8 & -4 & -20 & 16 & 20 & 0 & -28 \\ -2 & -36 & 40 & 20 & -50 & -20 & -26 \\ 72 & -6 & 0 & -66 & -6 & 30 & 4 \\ -14 & -4 & -6 & -12 & 28 & 18 & -56 \\ 28 & 6 & -18 & 8 & -2 & 16 & 2 \end{pmatrix}.$$

Thus this display, together with the fixed monomial order, specifies every coefficient, including zeros. Alternatively, the determinant of the displayed four quadrics is its coordinate-free recipe.

For any list of homogeneous forms $f_1, \dots, f_s$ define the degree- d Macaulay coefficient matrix $\mathcal{M}_d(f)$ as follows. Rows are all degree- d monomials in the same descending lexicographic order. Columns are the forms $x^\beta f_i$, with i outermost and all β of degree $d - \deg f_i$ in descending lexicographic order innermost. The entry in row α is the coefficient of x^α . All entries for this example are integers.

Matrix	Size	Certified rank	Minor modulo 101
$[F, G_{01}, \dots, G_{36}]$	35×25	25	41
$\mathcal{M}_5(q_0, q_1, q_2, q_3)$	56×80	56	44
$\mathcal{M}_9(F_{x_0}, F_{x_1}, F_{x_2}, F_{x_3})$	220×336	220	79

The differential minor consists of the first 25 rows and all columns; its full integer value is (6.4). The following deterministic rule specifies the other two minors without printing large matrices. Reduce the displayed coefficient matrix modulo 101. Scan columns from left to right. At the next pivot step choose the first remaining row with nonzero entry, exchange it with the next pivot row, normalize that row, and eliminate below. Skip a column if there is no nonzero remaining entry. Record the original row indices in the order selected and the pivot columns. The square submatrix of the *original*, unreduced matrix with exactly these row and column orders has the residue in the final column of the table. Full recorded index lists, along with the coefficient-generating program, are included in `evidence/K3_CERTIFICATES.json` and `checks/exact_k3.py`.

Every step is finite-field arithmetic on integer data. A nonzero residue of a specified square minor proves that integer minor is nonzero, so it proves a rank statement over $\mathbb{Q}$, not merely a rank in one numerical specialization. The Macaulay rank statements imply the homogeneous ideal containments used in Lemma 6.1. The exact integer determinant is also recomputed independently with rational-domain elimination. No floating-point rank, tolerance, or unverified smoothness inference is used.

For the corank-two quotient in (7.1), one may reproduce its quotient map by multiplying the six-dimensional quadratic quotient by

$$\begin{pmatrix} 0 & -1/2 & -1 & 1 & 0 & 0 \\ 0 & -1/2 & -1 & 0 & 1 & 0 \\ 0 & -3/2 & -2 & 0 & 0 & 1 \end{pmatrix}.$$

This matrix annihilates the three columns of $\text{Sym}^2 H$ and has rank three. It yields the displayed B_H, C_H and the quartic (7.2) by ordinary multiplication. The factor 2 in the coefficient of uv is essential: it records the square of $ux_2 + vx_3$, not a divided-power convention.

An additional exact regression test uses the normalized matrix (7.3) with

$$C = \begin{pmatrix} 0 & 1 & 0 \\ 0 & 0 & 0 \\ 0 & 1/2 & -3 \end{pmatrix}.$$

Then $h(u, v) = uv(u-v)(u-2v)$, so the four ideals are (a, b) , (c, d) , $(a-c, b-d)$, and $(a-2c, b-2d)$. Elimination computes both sides of (1.10) over $\mathbb{Q}$, and mutual Gröbner reduction verifies their equality. It also checks that δ^2 is nonzero, that every element of $\mathfrak{m}\delta^2$ vanishes, and that $\delta^3 = 0$. This test checks a concrete instance; the general coefficient-ring proof is Lemma 7.4 and Theorem 7.5, not an extrapolation from that instance.

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